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CAPSTONE PROJECT BY TEAM A
A DATA SCIENCE APPROACH TO FORECAST ELECTRICITY CONSUMPTION IN AUSTRALIA

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Abstract

Accurate electricity demand forecasting is critical for efficient energy management, cost optimisation, and grid stability. Traditional statistical approaches such as SARIMAX have been widely used for short-term forecasting, while machine learning and transformer-based models have emerged as promising alternatives for capturing complex temporal dependencies.

This study evaluates the performance of SARIMAX, Gradient Boosting (GB), LSTM, and a transformer-based model (PatchTST) using half-hourly electricity demand data from New South Wales, Australia. Models are assessed across multiple forecasting horizons (48, 336, and 720 steps) using RMSE and MAE under a consistent rolling-window evaluation framework.

For the 48-step horizon, PatchTST (RMSE $\approx$ 468) performed the best, followed by Gradient Boosting (RMSE 537), SARIMAX (RMSE $\approx$ 545), and LSTM (RMSE $\approx$ 565). This result is continued for 336-step horizon but with higher RMSE values: 637, 651, 656 and 791 respectively for PatchTST, SARIMAX, Gradient Boosting and LSTM. For longer horizons (720-steps), SARIMAX achieved the best performance (RMSE $\approx$ 668), followed by Gradient Boosting (RMSE $\approx$ 691), PatchTST (RMSE $\approx$ 695) and LSTM (RMSE $\approx$ 859).

The results reveal a clear trade-off between model complexity and forecasting horizons. PatchTST demonstrated superior performance for short- and medium-term forecasting by effectively capturing complex short-range temporal dependencies. However, at the long-term horizon, the predictive signal within historical data degraded. In this highly uncertain regime, the simpler SARIMAX model marginally outperformed the more advanced architectures, proving it more robust against noise. Gradient Boosting offered stable but intermediate accuracy across all horizons, while LSTM consistently underperformed due to overfitting and instability over extended sequences.

Ultimately, the findings demonstrate that no single architecture is universally optimal: advanced transformer models excel when deterministic temporal signals are strong, whereas classical statistical baselines remain highly relevant for long-range predictions, contrary to what was believed. Furthermore, achieving significant gains in long-term forecasting will likely require the integration of exogenous variables, such as weather data, rather than strictly increasing model complexity.

Contents

1	Introduction.....	1
2	Literature Review	2
3	Material and Methods.....	4
3.1	Software	4
3.2	Description of the Data	4
3.3	Pre-processing Steps.....	5
3.4	Data Cleaning.....	5
3.5	Assumptions	6
4	Exploratory Data Analysis.....	7
4.1	Overview	7
4.2	Temporal Patterns in Electricity Demand	7
4.3	Relationship Between Temperature and Electricity Demand.....	7
4.4	Forecast Demand vs Actual Demand	9
4.5	Correlation Analysis and lagged relationship.....	10
4.6	Seasonal and Daily Demand Patterns	13
4.7	Seasonal Decomposition Using MSTL	14
4.8	Distribution and Outlier Analysis	16
4.9	Key Findings from EDA	17
5	Modelling.....	19
5.1	Modelling Methods.....	19
5.2	Tuning of Hyper-parameters.....	21
5.3	Model Training Behaviour	25
6	Results	29
6.1	Summary of Results.....	29
7	Discussion	32
8	Conclusion and Further Issues	33
8.1	Summary of Study	33
8.2	Key Findings.....	33
8.3	Implications	33
8.4	Future Work	34
9	Codes	37
10	Appendix.....	37

1 Introduction

Electricity demand forecasting is essential for the efficient operation and planning of modern power systems. Accurate forecasts allow energy providers to optimise generation, reduce operational costs, and maintain system reliability. Conversely, poor predictions, particularly during peak demand periods, can result in significant economic losses or supply disruptions. Developing robust and reliable forecasting models is therefore a critical challenge in contemporary energy management.

This project focuses on forecasting electricity demand in New South Wales (NSW), Australia, using historical demand, temperature, and forecast data. Electricity demand is influenced by a range of factors, including weather conditions, seasonal patterns, and human activity. Among these, temperature plays a particularly significant role due to its influence on heating and cooling requirements, making it a key predictor in demand modelling.

To address this problem, a range of modelling approaches is explored, spanning both traditional statistical techniques and modern machine learning methods. Specifically, the study implements a Seasonal Autoregressive Integrated Moving Average model with exogenous variables (SARIMAX), a Gradient Boosting model, a Long Short-Term Memory (LSTM) neural network, and a Patch Time Series Transformer (PatchTST). These models are evaluated and compared based on their forecasting accuracy and their ability to capture complex temporal dynamics.

The objectives of this study are threefold. First, to characterise relationships within the electricity demand dataset through exploratory data analysis. Second, to develop and evaluate multiple forecasting models across different methodological frameworks, with a focus on multi-horizon prediction and how forecast accuracy degrades with increasing lead time. Third, to compare model performance and identify the most effective approach for electricity demand prediction in this context.

The outcomes of this project provide insight into the relative strengths and limitations of different forecasting techniques and demonstrate the potential of advanced machine learning models to improve electricity demand prediction.

2 Literature Review

Electricity demand forecasting has been extensively studied due to its importance in power system operation and planning. Traditional statistical time series models such as Autoregressive Integrated Moving Average (ARIMA) and its seasonal variant (SARIMA) have long been used to model electricity demand due to their ability to capture trends and recurring seasonal patterns. These approaches have demonstrated strong performance in short-term forecasting tasks where demand follows predictable daily and weekly cycles (Box et al., 2015; Taylor, 2003). Their relative simplicity, interpretability, and low computational requirements make them a reliable baseline for comparison. Extensions of these models have also been applied to longer horizons, with studies showing that incorporating multiple seasonal components can improve forecasting accuracy over extended periods (Fan & Hyndman, 2012).

While traditional statistical models provide a strong baseline for electricity demand forecasting, more flexible machine learning approaches have been introduced to better capture complex, non-linear relationships in the data. One such approach is gradient boosting, an ensemble learning technique that builds models sequentially by fitting decision trees to the residual errors of previous models (Friedman, 2001). This iterative process allows gradient boosting to minimise prediction errors effectively and achieve high accuracy. Compared to individual models, ensemble methods such as gradient boosting improve robustness and generalisation performance (Hastie, Tibshirani and Friedman, 2009). Modern implementations, including XGBoost and LightGBM, further enhance performance through regularisation and computational efficiency, making them suitable for large-scale forecasting problems (Chen and Guestrin, 2016; Ke et al., 2017).

In time series forecasting applications such as electricity demand prediction, gradient boosting models are particularly effective when combined with feature engineering techniques such as lagged variables and exogenous inputs like temperature. Unlike traditional statistical approaches, these models do not rely on assumptions of linearity or stationarity, allowing them to capture more complex patterns in the data. Empirical studies have shown that tree-based ensemble methods can outperform classical models when sufficient data and relevant features are available (Makridakis et al., 2018; Taieb and Hyndman, 2014). However, while gradient boosting provides strong predictive performance, it may require careful tuning and feature design to achieve optimal results.

More recently, deep learning approaches have been introduced to better capture nonlinear relationships and complex temporal dependencies in electricity demand data. Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, are particularly well suited to time series forecasting due to their ability to retain information over long sequences using gated memory cells (Hochreiter & Schmidhuber, 1997). In the context of electricity demand, LSTM models have shown strong performance when incorporating exogenous variables such as temperature, as well as lagged demand and calendar features. Studies such as Kong et al. (2017) and Marino et al. (2016) demonstrate that LSTMs can outperform traditional statistical models in short-term forecasting by effectively learning nonlinear demand patterns and temporal dependencies.

Despite their advantages, LSTM models have limitations when applied to medium- and long-term forecasting horizons. While they are capable of modelling long sequences, their predictive accuracy tends to decrease as the forecast horizon increases, as longer-term forecasts depend more heavily on stable trends and seasonal structures rather than short-term temporal relationships. Research such as Peřka and Dudek (2020) shows that hybrid or decomposition-based approaches, which combine statistical models with neural networks, can improve performance in these cases. This highlights the continued importance of classical methods and feature engineering when addressing forecasting horizons beyond short-term applications.

Transformer-based architectures have emerged as a newer approach for time series forecasting, offering an alternative to recurrent models by leveraging attention mechanisms to capture dependencies across long sequences. However, early work questioned their effectiveness in this domain, with studies showing that simpler statistical and machine learning models could outperform transformer-based approaches for long-term forecasting tasks (Li et al., 2022; Zeng et al., 2022). More recent developments, such as PatchTST (Nie et al., 2023), have addressed some of these limitations by introducing techniques such as patching and channel independence, improving performance on long-horizon forecasting benchmarks. These advancements suggest that transformer-based models may be more competitive when appropriately adapted for time series data.

While prior studies show LSTM performs well in short-term forecasting, its performance degrades over longer horizons. Although transformer-based models have shown mixed results, recent architectures such as PatchTST suggest improvements in long-horizon forecasting. However, empirical comparisons across short-to-medium horizons (48–720 timesteps) remain limited, particularly in the Australian electricity market context.

Overall, the literature suggests that model performance in electricity demand forecasting is strongly dependent on the forecast horizon. Traditionally, it is assumed that statistical models remain highly effective for short-term and structured seasonal forecasting, while LSTM models provide advantages in capturing nonlinear and short-term temporal dependencies. Conversely, recent studies claim transformer-based models show promises for long-term forecasting but require careful design and sufficient data to outperform simpler approaches. PatchTST introduces patching and channel independence to improve long-horizon forecasting performance and reduce overfitting. In addition, prior studies such as Nie et al. (2023) primarily benchmark PatchTST against deep learning and linear machine learning models, with limited comparison to classical statistical models such as SARIMAX or recurrent models such as LSTM. This study addresses this gap by evaluating these model families under a consistent forecasting framework, testing the assumption that advanced architectures inherently outperform classical methods at longer horizons.

3 Material and Methods

3.1 Software

A range of software tools was utilised to support collaboration, model development, and reporting throughout this project.

Microsoft Teams served as the primary platform for group communication and coordination. Its integration of messaging, video conferencing, and file sharing enabled efficient collaboration, while features such as meeting recording and live captions supported documentation and review of key discussions. Integration with OneDrive also provides centralised file storage and version control.

GitHub was adopted for collaborative code management and version control. It enables tracking of all code changes, supports branching and pull requests, and allows multiple team members to work on different components simultaneously without conflicts. The use of issues and task tracking features also supported project management, including assigning responsibilities and monitoring progress.

Python was the primary programming language for data analysis and modelling, selected for its extensive ecosystem of scientific libraries. Pandas and NumPy were used for data preprocessing, scikit-learn for machine learning models such as Gradient Boosting, statsmodels for SARIMAX implementation, and PyTorch for deep learning models including LSTM and PatchTST. This environment enabled efficient end-to-end workflows, from data preparation through to model evaluation.

Jupyter Notebooks were used to support reproducible analysis by combining code, outputs, and explanatory text within a single document. This facilitated transparency and ease of review. The final report was compiled in Microsoft Word to ensure a structured presentation and compliance with submission requirements.

This combination of tools supported reproducible model development and enabled consistent evaluation of forecasting performance across multiple prediction horizons.

3.2 Description of the Data

The dataset used in this study consists of three primary data sources related to electricity demand in New South Wales (NSW): total electricity demand, temperature observations, and forecast demand.

All datasets were stored as compressed CSV files (.csv.zip) within a structured GitHub repository. In total, four raw data files were used: one file for total demand, one for temperature, and the forecast demand dataset split into two parts due to its large size.

The forecast demand data was stored as two separate files, `forecastdemand_nsw.csv.zip.parta` (100 MB) and `forecastdemand_nsw.csv.zip.partb` (14 MB), which were combined programmatically into a single compressed file prior to analysis. The temperature dataset (`temperature_nsw.csv.zip`) has a size of approximately 899 KB, while the total demand dataset (`totaldemand_nsw.csv.zip`) is approximately 1.24 MB in size.

The total demand dataset contains 196,513 observations with three variables: DATETIME, TOTALDEMAND, and REGIONID. The temperature dataset contains 220,326 observations with variables LOCATION, DATETIME, and TEMPERATURE. The forecast dataset is significantly larger, containing over 10 million observations and multiple forecast-related variables.

To facilitate analysis, the datasets were later merged into a single unified dataset based on the DATETIME variable, ensuring alignment across all three data sources.

The large size of the forecast dataset (over 10 million observations) provides high temporal resolution, which is particularly advantageous for evaluating model performance across different forecast horizons. The inclusion of both temperature and forecast demand variables enables investigation of both exogenous drivers and existing operational forecasts.

3.3 Pre-processing Steps

Several preprocessing steps were performed to transform the raw datasets into a format suitable for modelling.

First, the forecast dataset was reconstructed by combining the two split files into a single compressed file. This step was necessary due to file size constraints in storage and version control systems.

Next, all DATETIME fields were converted into a consistent datetime format. Different parsing formats were required for different datasets to ensure accurate conversion. This step was critical for enabling time-based alignment and preventing errors during merging.

Following datetime standardisation, all datasets were sorted chronologically to preserve temporal order, which is critical for sequential models such as LSTM and transformer-based architectures. It is also equally important for SARIMAX because of its autoregressive and moving average terms.

The forecast dataset originally contained multiple forecast entries for each timestamp. To reduce redundancy and ensure consistency, the data was aggregated by computing the mean FORECASTDEMAND for each DATETIME value. This transformation reduces noise arising from multiple forecast entries while retaining the central tendency of the forecast signal, ensuring consistency when comparing model predictions across different horizons.

The datasets were then merged using an inner join on the DATETIME variable. This ensured that only observations with complete information across all datasets were retained. The resulting dataset contains 195,947 aligned observations.

Additionally, categorical variables such as REGIONID and LOCATION were removed, as they contained only a single unique value and therefore did not contribute meaningful variation for modelling.

3.4 Data Cleaning

Data cleaning procedures were implemented to ensure data quality and consistency prior to modelling.

Missing data primarily arose from datetime parsing issues. These were addressed by converting all DATETIME fields using appropriate formats and removing any rows where conversion failed (i.e., rows with invalid timestamps). This ensured that all remaining observations had valid and consistent time indices.

Furthermore, the use of an inner join during dataset merging inherently removed any observations with missing values across the datasets. As a result, the final merged dataset contains no missing values in the key variables TOTALDEMAND, TEMPERATURE, and FORECASTDEMAND.

Outlier detection was conducted using box plots for both electricity demand and temperature

variables. While some extreme values were identified, these were retained in the dataset as they likely represent genuine real-world phenomena, such as peak electricity demand during extreme weather conditions, rather than data errors. Retaining extreme values is particularly important in electricity demand forecasting, as peak demand events are critical for system reliability and model evaluation.

Finally, the dataset was checked for consistency, including duplicate timestamps and alignment across variables, ensuring that the final dataset is clean, complete, and suitable for modelling.

3.5 Assumptions

Several assumptions were made throughout the data preparation and modelling process to ensure the feasibility and validity of the analysis.

It is assumed that the recorded electricity demand and temperature data are accurate and reliably measured, with no systematic measurement errors. Any extreme values observed in the dataset are assumed to reflect genuine real-world events, such as periods of unusually high or low demand, rather than data anomalies.

The datasets are assumed to be temporally aligned after preprocessing. The merging process relies on the DATETIME variable, and it is assumed that observations sharing the same timestamp across datasets correspond to the same real-world time period. Any minor discrepancies in timing are considered negligible after aggregation.

It is assumed that the removal of categorical variables such as REGIONID and LOCATION does not result in loss of meaningful information, as these variables contain only a single unique value and therefore do not contribute to model performance.

It is assumed that the historical patterns observed in the data, including seasonal and daily trends, are representative of future behaviour. This assumption underpins the validity of time series forecasting models such as SARIMAX, Gradient Boosting, and deep learning models.

It is also assumed that the relationship between temperature and electricity demand remains stable across the forecasting horizons considered, although this relationship may weaken for longer-term predictions.

Finally, it is assumed that there are no significant structural breaks in the data over time (e.g., major policy changes or disruptions) that would invalidate the learned relationships between variables.

4 Exploratory Data Analysis

4.1 Overview

The objective of this analysis is to identify key temporal patterns and relationships within the data that informs model selection and forecast performance across different prediction horizons. The EDA focuses on examining temporal trends, relationships between variables such as temperature and demand, and the performance of existing forecast demand data.

4.2 Temporal Patterns in Electricity Demand

Electricity demand exhibits strong and consistent temporal variation, as illustrated in the time series plot (Figure 4.1). Demand fluctuates between approximately 5,500 MW and 14,500 MW, reflecting substantial variability driven by both environmental and behavioral factors.

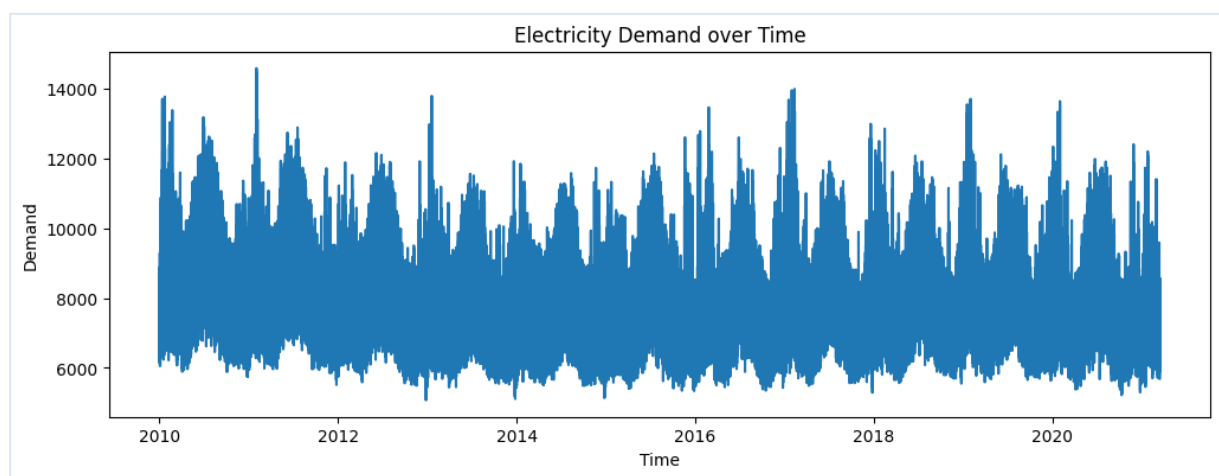


Figure 4-1: Demand in Megawatts (MW)

Clear periodic patterns are observed at multiple time scales. At a daily level, demand follows a repeatable cycle characterised by low consumption during early morning hours and peak demand in the evening. At a broader scale, seasonal variation is also evident, with higher demand during winter months and lower demand during milder periods.

These observations indicate that electricity demand is governed by multi-scale temporal structure, combining short-term (intra-day) and long-term (seasonal) dependencies. This has important implications for model selection.

First, the presence of strong daily periodicity suggests that lagged features at 24-hour intervals (or equivalent timesteps) will be important predictors. Second, the longer-term seasonal variation indicates the need to capture annual trends, either through explicit seasonal terms (e.g., SARIMAX) or learned representations (e.g., neural networks).

Finally, the persistence and regularity of these patterns suggest that models capable of capturing temporal dependencies—such as recurrent neural networks (LSTM) or transformer-based architectures—are well suited to this task. Classical time series models may also perform well if appropriately configured to incorporate both seasonal and autoregressive components.

4.3 Relationship Between Temperature and Electricity Demand

The relationship between temperature and electricity demand was examined using a scatter plot (Figure 4.2), revealing a clear non-linear, U-shaped pattern.

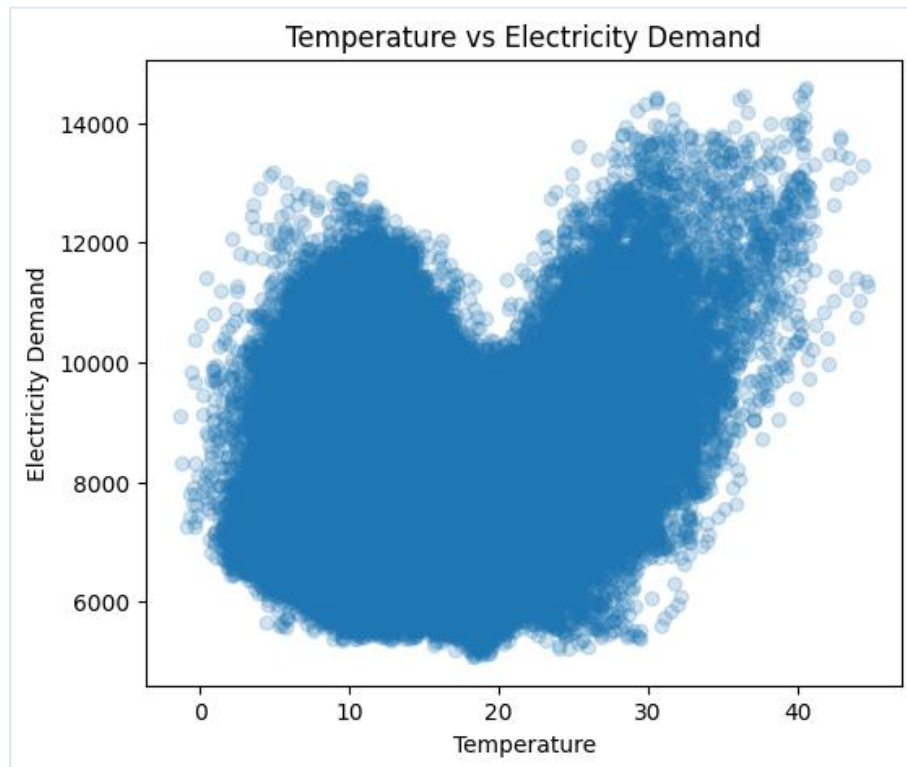


Figure 4-2: Demand in Megawatts (MW), Temperature in Celsius (C)

Electricity demand increases at both low and high temperature extremes. At low temperatures, increased demand is driven by heating requirements, while at high temperatures, demand rises due to air-conditioning usage. Demand is lowest at moderate temperatures, where minimal climate control is required.

This behaviour indicates that temperature influences demand in a **non-linear and asymmetric manner**, which cannot be fully captured by a simple linear relationship. As a result, models that assume linear dependence between predictors and response—such as a naïve SARIMAX specification—may fail to accurately represent this effect.

However, this limitation is not inherent to classical models, but rather to the choice of feature representation. By introducing non-linear transformations of temperature (e.g., quadratic terms, spline functions, or piecewise variables), traditional time series models can be extended to capture this relationship effectively.

Alternatively, machine learning and deep learning models—such as gradient boosting methods or neural networks—can learn this non-linear relationship directly from the data without requiring explicit feature engineering.

These findings highlight the importance of appropriately modelling temperature effects, either through engineered features in statistical models or through flexible function approximation in machine learning approaches.

4.4 Forecast Demand vs Actual Demand

The relationship between forecast demand and actual demand was examined using a scatter plot (Figure 4.3), which shows a strong positive association. This indicates that the existing forecasting system is generally effective at capturing the overall trend in electricity demand.



Figure 4-3: Provided forecast vs Actual Demand In Megawatts (MW) for each of the 48 forecast horizon steps

However, closer inspection reveals that forecast accuracy varies across different demand levels. Greater dispersion is observed at higher demand values, indicating reduced accuracy during peak demand periods.

This suggests the presence of **heteroscedastic error**, where prediction errors increase with demand magnitude. Such behaviour is common in electricity demand forecasting, as extreme demand conditions are often driven by complex and non-linear interactions between temperature, human activity, and system constraints.

The reduced accuracy at peak demand is particularly important, as these periods are typically of greatest operational and economic significance. Underestimation or overestimation during peak periods can have substantial implications for system reliability and cost.

These findings indicate that while the baseline forecasting approach captures general trends effectively, it struggles to model extreme conditions. This highlights the need for more flexible modelling approaches that can better capture non-linear behaviour and regime-dependent dynamics, particularly during periods of high demand.

4.5 Correlation Analysis and lagged relationship

To quantify the relationship between temperature and electricity demand, an initial correlation analysis was performed using a standard correlation matrix (Figure 4.4). The results indicate only a weak linear relationship between temperature and demand.

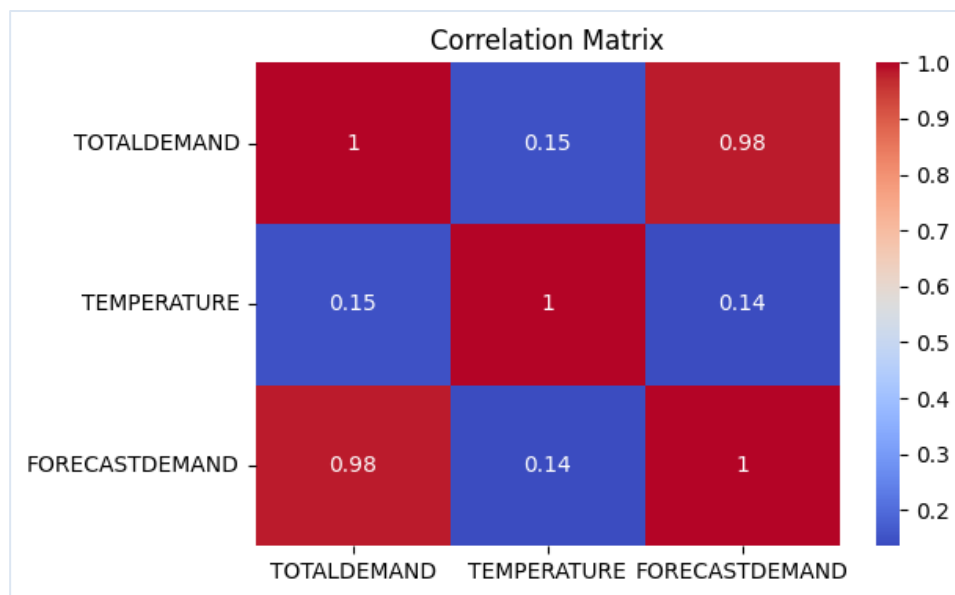


Figure 4-4. Correlation matrix of total demand, temperature and actual forecast demand.

However, this finding is misleading, as it does not account for the temporal structure of the data. Electricity demand does not respond instantaneously to changes in temperature, and the relationship between the two variables is both time-dependent and non-linear. As a result, zero-lag correlation alone is insufficient to capture the true relationship.

The autocorrelation function (ACF) over short lags, shown in Figure 4.5, reveals a strong and gradually decaying correlation structure. The autocorrelation is close to one at lag 0 and remains significantly positive across multiple subsequent lags, decreasing slowly rather than abruptly. This indicates that electricity demand is highly persistent, with current values strongly influenced by recent observations.

Unlike the partial autocorrelation function (PACF), which isolates direct relationships, the ACF captures both direct and indirect dependencies. The slow decay observed in the ACF suggests that the time series contains a combination of autoregressive and seasonal components rather than being purely random or short-memory. This persistence implies that models must account for temporal continuity and cannot rely solely on short-term dependencies.

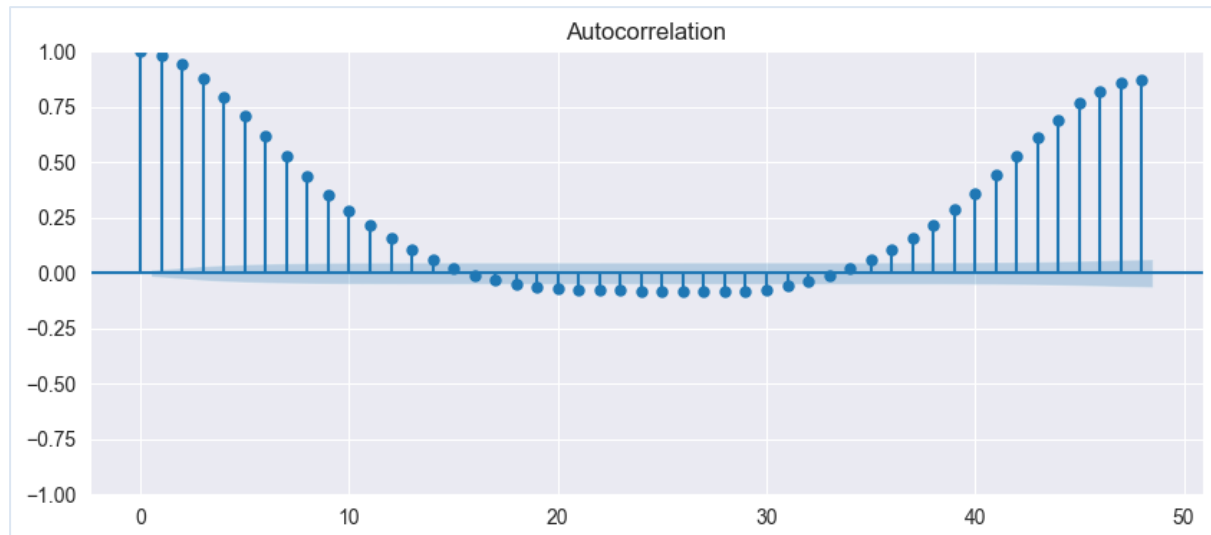


Figure 4-5: Y-axis is correlation, X-axis is timesteps

The PACF, shown in Figure 4.6, provides insight into the direct relationship between the time series and its lagged values after accounting for intermediate lags. The PACF exhibits strong spikes at the first few lags, particularly at lag 1, indicating a strong immediate dependence of electricity demand on its most recent values. Beyond these initial lags, the partial autocorrelations quickly diminish and fluctuate around zero, suggesting that the direct influence of additional past observations is limited once short-term effects are accounted for.

This pattern is characteristic of a process that can be reasonably approximated using a low-order autoregressive structure. The sharp drop-off after the first few lags supports the selection of a small autoregressive order in models such as SARIMAX, as higher-order terms are unlikely to provide substantial additional explanatory power.

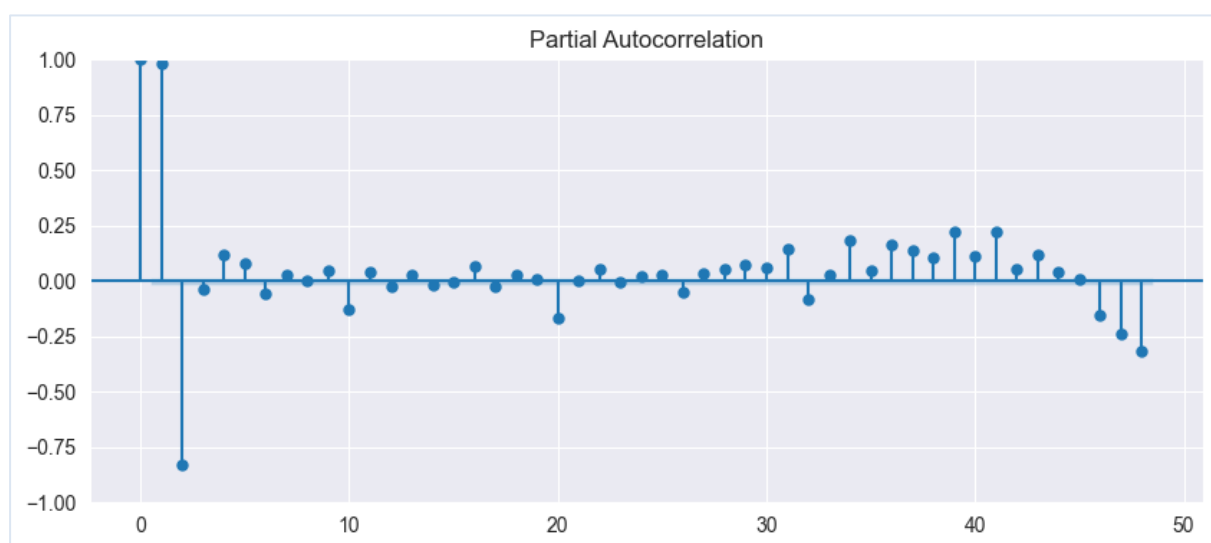


Figure 4-6: Y-axis is partial correlation, X-axis is timesteps

The extended autocorrelation function, shown in Figure 4.7, provides clear evidence of strong periodic

behaviour in electricity demand. Distinct repeating peaks are observed at regular intervals, corresponding to approximately 48 lags, which aligns with a 24-hour cycle in half-hourly data. This repeating sinusoidal pattern indicates the presence of a dominant daily seasonality in the data.

In addition to the primary daily cycle, the persistence of this pattern across multiple periods suggests stable and consistent temporal structure over time. The alternating positive and negative correlations between peaks further reinforce the cyclical nature of the series. This behaviour confirms that electricity demand is not only autocorrelated but also strongly seasonal.

A lagged correlation analysis was then conducted (Figure 4.8) to examine how temperature influences demand across different time offsets. The results show that the strongest positive correlation occurs at a lag of approximately one hour, indicating that electricity demand responds to temperature changes with a short delay. This behaviour is consistent with real-world dynamics, where thermal inertia in buildings and delayed activation of heating and cooling systems introduce a lagged demand response.

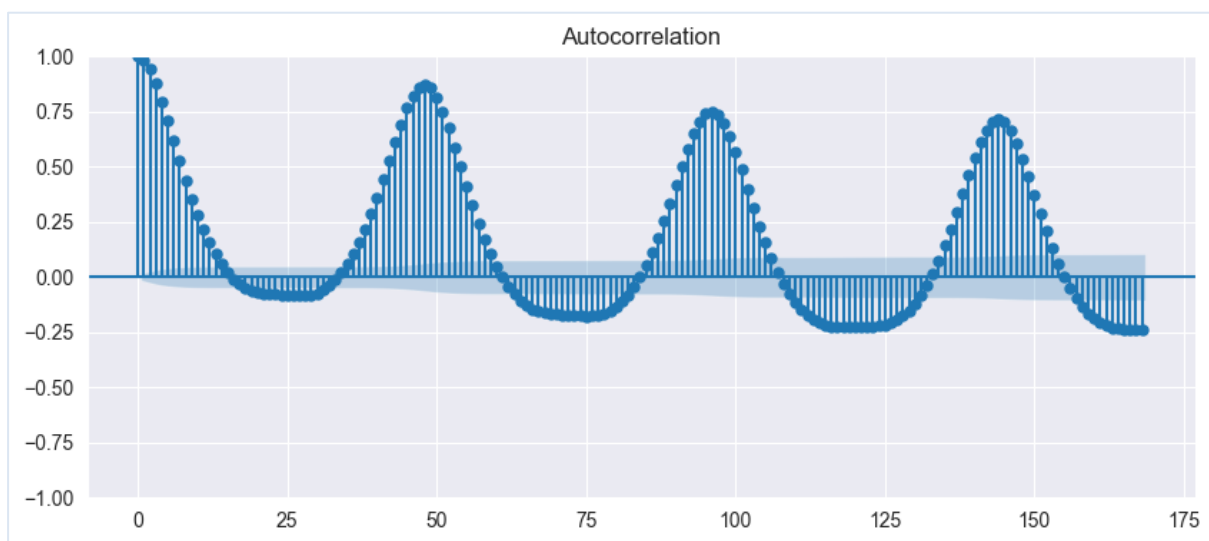


Figure 4-7: Y-axis is correlation, X-axis is timesteps

A pronounced negative correlation is observed at lags of approximately 24–30 intervals (12–15 hours), reflecting the diurnal cycle. Periods of high daytime temperature and demand are followed by lower nighttime values, producing an inverse relationship at these offsets.

The overall sinusoidal pattern of the lagged correlation further confirms the presence of strong periodic behaviour in the data. Importantly, this structure is not captured by the zero-lag correlation matrix, highlighting the limitations of conventional correlation analysis for time series data.

These findings demonstrate that the relationship between temperature and demand is inherently multi-scale, involving both short-term lag effects (≈ 1 hour) and longer-term periodic dependencies (≈ 24 hours). This has direct implications for modelling. Effective forecasting approaches should incorporate lagged features and temporal structure, either explicitly through autoregressive terms (e.g. SARIMAX) or implicitly through sequence-based models such as LSTMs and transformer-based architectures.

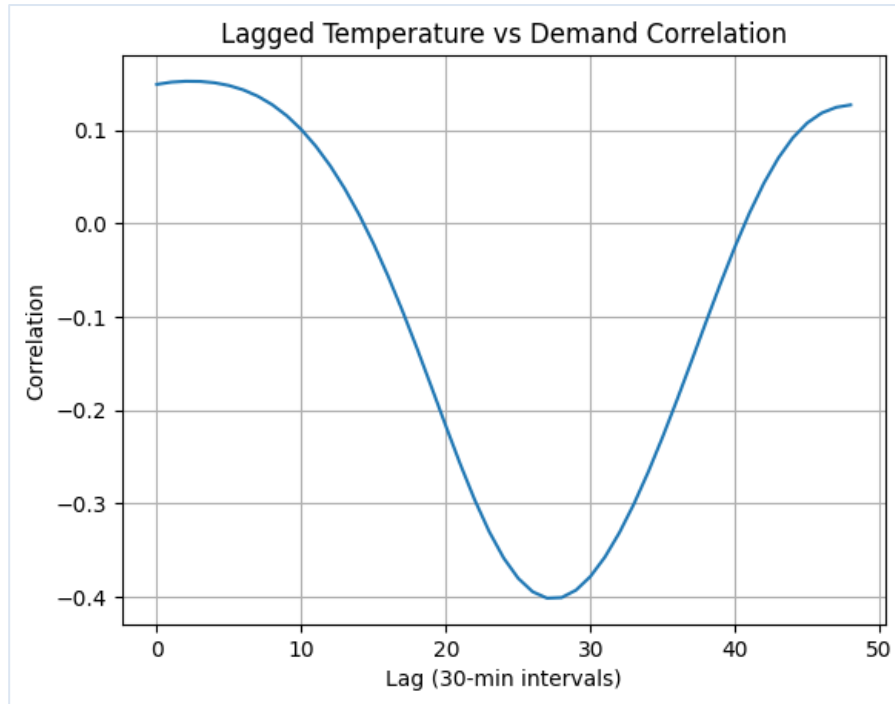
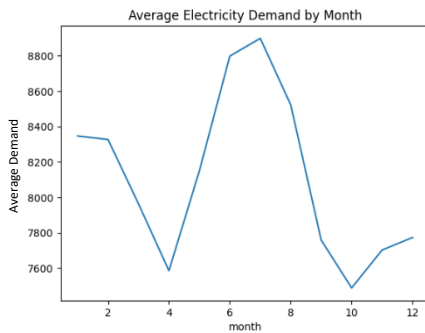


Figure 4-8: Y-axis is correlation, X-axis is timesteps in 30-min lag intervals

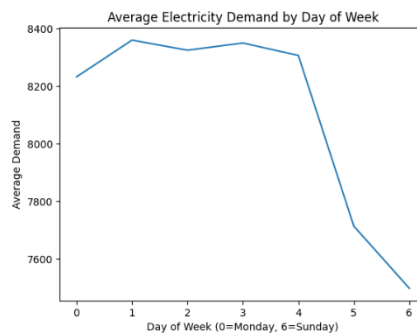
4.6 Seasonal and Daily Demand Patterns

To further characterise the temporal structure of electricity demand, patterns were analysed across multiple time scales, including monthly, weekly, and hourly variations (Figures 4.9a–4.9c)

Monthly Patterns



Day-of-Week Patterns



Hourly Patterns

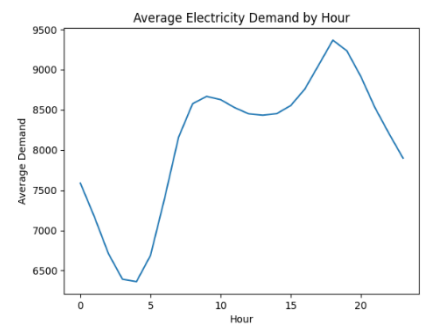


Figure 4-9. Plots showing the seasonal demand patterns for Monthly (left), Day of the week (Middle) and Hourly (right).

At the monthly level, demand exhibits clear seasonal variation, with higher consumption during winter months (June to August) and lower demand during milder periods such as April and October. This reflects increased heating requirements during colder conditions and reduced energy usage during transitional seasons.

At the weekly level, demand is consistently lower on weekends compared to weekdays, with Sunday showing the lowest average demand. This pattern is likely driven by reduced industrial and commercial activity, highlighting the influence of human behavioural cycles on electricity consumption.

At the hourly level, demand follows a highly regular daily cycle. The lowest demand occurs during early morning hours (approximately 3–5 a.m.), followed by a steady increase throughout the day and a peak in the evening (around 6–7 p.m.). This reflects typical daily activity patterns, including residential

energy use after working hours.

Taken together, these results demonstrate that electricity demand is influenced by distinct but interacting temporal components, spanning seasonal (monthly), behavioural (weekly), and diurnal (hourly) cycles.

This decomposition has important implications for modelling. Each time scale represents a different source of variation and should be incorporated appropriately. Seasonal effects may be captured using periodic terms (e.g. annual seasonality in SARIMAX), while weekly and hourly patterns can be represented through categorical variables or cyclic encodings (e.g. sine and cosine transformations).

For machine learning and deep learning models, these temporal patterns may be learned implicitly; however, providing structured time-based features can still improve model performance and convergence.

4.7 Seasonal Decomposition Using MSTL

To further analyse the temporal structure of electricity demand, a decomposition was performed using Multiple Seasonal-Trend decomposition using LOESS (MSTL) (Bandara et al., N.D.), which separates the time series into trend, seasonal, and residual components across multiple seasonal periods.

The decomposition reveals distinct components corresponding to daily, weekly, and longer-term seasonal patterns, as shown in Figures 4.10a, 4.10b.

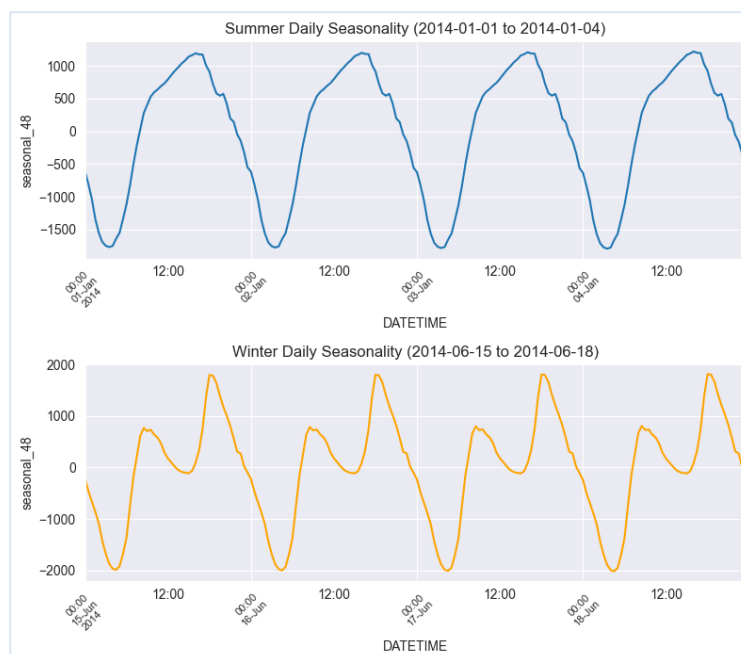


Figure 4-10 a) summer daily seasonality, b) winter daily seasonality

The daily seasonal **component** (Figure 4.10) exhibits a consistent intra-day structure across both summer and winter periods, characterised by low demand during early morning hours and a peak in the evening. While the overall shape remains stable between seasons, the amplitude of the cycle differs significantly. In particular, the winter period shows a more pronounced evening peak and deeper troughs compared to summer, indicating increased variability in demand during colder months.

The weekly seasonal component (Figure 4.11) reflects clear behavioural patterns, with reduced demand observed during weekends, particularly on Sundays. This pattern is consistent across both summer and winter; however, the magnitude of the weekend reduction is greater during winter, suggesting a stronger dependence on external conditions during colder periods.

The trend component (Figure 4.12) captures longer-term seasonal variation, with demand generally increasing during winter months and decreasing during milder periods. This aligns with increased energy usage associated with heating demand.

Overall, the decomposition demonstrates that electricity demand can be represented as a combination of multiple seasonal processes operating at different time scales. Importantly, while the temporal structure of these components remains relatively stable, their magnitude varies across seasons.

This has important implications for modelling. It suggests that effective forecasting models must not only capture periodic structure, but also allow for season-dependent variation in amplitude. Classical models such as SARIMAX may require additional seasonal terms or exogenous variables to capture this behaviour, while machine learning and deep learning models may be better suited to learning these interactions directly from the data.

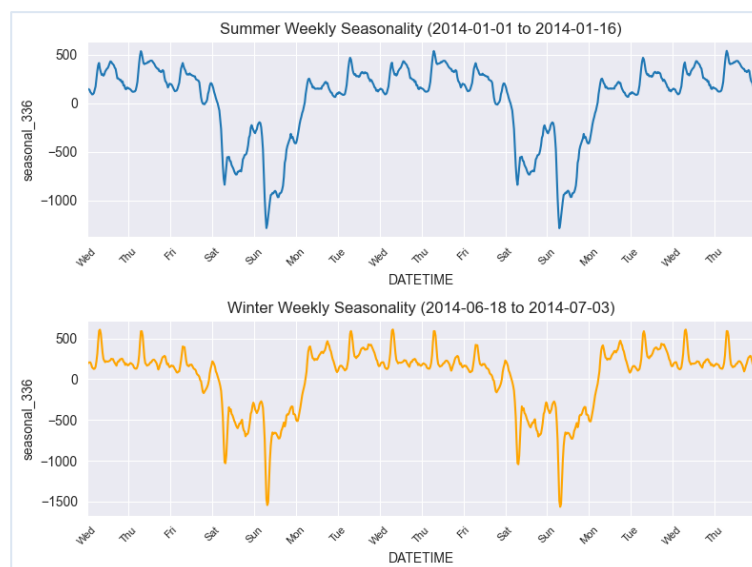


Figure 4-11. Seasonal demand variation for summer and winter.



Figure 4-12. Monthly demand trend.

4.8 Distribution and Outlier Analysis

The distribution of total electricity demand was analysed to assess its statistical properties and suitability for modelling. The raw demand distribution (Figure 4.13) exhibits a noticeable right skew, with a long tail extending toward higher demand values. This indicates that while most observations fall within a moderate range, extreme high-demand events occur less frequently but contribute significantly to overall variability.

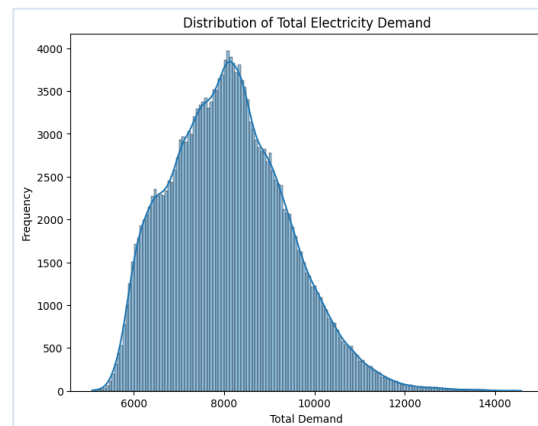


Figure 4-13. Histogram of total electricity demand.

To address this skewness, a logarithmic transformation was applied. The resulting distribution (Figure 4.14) is substantially more symmetric and closely resembles a normal distribution. This transformation reduces the influence of extreme values and stabilises variance, which is particularly beneficial for classical statistical models such as SARIMAX that assume approximately normally distributed residuals.

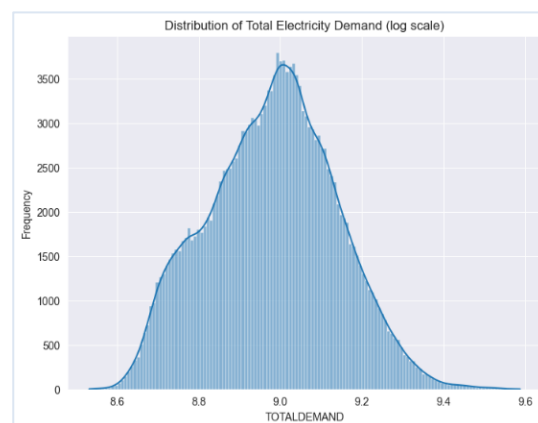


Figure 4-14. Histogram of the log transformed total electricity demand.

The presence of extreme values is further illustrated in the box plots of electricity demand and temperature (Figure 4.15). Both variables exhibit clear outliers, particularly at higher values. Importantly, these observations are not likely to represent noise or measurement error but rather correspond to real-world extreme events such as heatwaves or cold spells, which drive unusually high demand.

The combination of right-skewness and visible outliers suggests that electricity demand exhibits heteroscedastic behaviour, where variability increases with demand magnitude. This finding is consistent with earlier observations (Section 4.4), where forecast errors were shown to increase during peak demand periods.

These characteristics have important implications for modelling. Logarithmic transformation or other

variance-stabilising approaches may improve the performance of classical time series models, while machine learning and deep learning models must be robust to extreme values to ensure reliable performance during high-demand conditions.

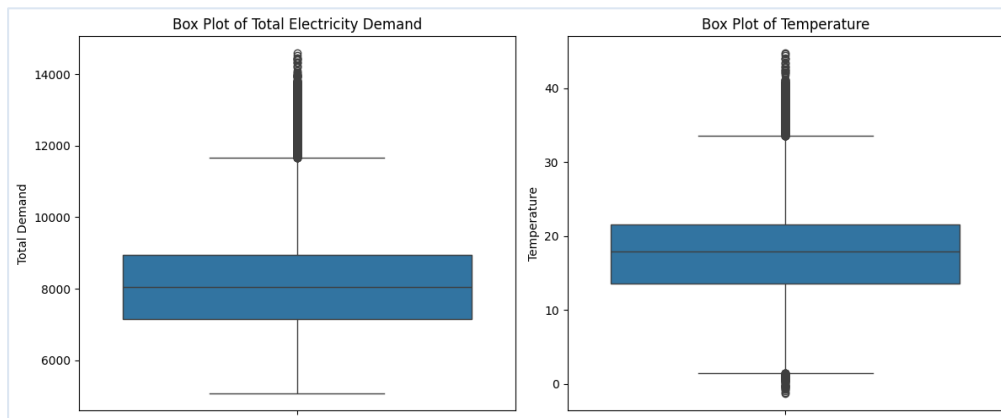


Figure 4-15. Box plots of total demand (left) and temperature (right).

4.9 Key Findings from EDA

The exploratory data analysis reveals several key characteristics of electricity demand that have direct implications for forecasting model design.

First, electricity demand exhibits strong multi-scale temporal structure, including daily, weekly, and seasonal cycles. These patterns are highly regular and persistent, indicating that temporal dependencies are a dominant driver of demand and must be explicitly captured in any forecasting model.

Second, the relationship between temperature and demand is non-linear and asymmetric, with increased demand observed at both low and high temperature extremes. This behaviour cannot be adequately captured by simple linear relationships and requires either feature transformation or flexible modelling approaches.

Third, lagged correlation analysis demonstrates that demand responds to temperature with a short delay (approximately 1 hour), while also exhibiting longer-term periodic dependencies. This confirms that the relationship between key variables is time-dependent, requiring models capable of capturing lagged effects.

Fourth, analysis of forecast performance shows that prediction errors increase significantly during periods of high demand. This indicates the presence of heteroscedasticity and highlights that extreme demand conditions are more difficult to model, despite being of greatest operational importance.

Fifth, seasonal decomposition reveals that while temporal patterns are structurally consistent, their amplitude varies across seasons. This suggests that models must capture not only periodic behaviour, but also season-dependent variation in magnitude.

Finally, the distribution of demand exhibits slight skewness and contains extreme values associated with real-world events such as heatwaves and cold spells. These observations reinforce the need for robust modelling approaches that can handle variability and extreme conditions effectively.

Taken together, these findings indicate that an effective forecasting model must:

- Capture multi-scale temporal dependencies
- Model non-linear relationships between variables
- Incorporate lagged effects
- Handle heteroscedasticity and extreme values
- Adapt to seasonal variation in demand magnitude

These requirements motivate the use of advanced modelling approaches, including machine learning and deep learning methods, which are capable of learning complex, non-linear, and time-dependent relationships directly from the data.

5 Modelling

5.1 Modelling Methods

5.1.1 Framework

A key focus of this study is to evaluate how different modelling techniques compare to each other. Specifically, four modelling approaches were implemented and their performance analysed: SARIMAX (statistical model), Gradient Boosting (GB), Long Short-Term Memory (LSTM), and PatchTST (transformer-based model). We evaluate the effectiveness of these models on a framework frequently seen in the research community mentioned in chapter 2 (Nie et al., 2023; Li et al., 2022; Zeng et al., 2022). The framework is described below:

- 48, 336, and 720 steps, corresponding to short-, medium-, and long-term forecast horizons
- Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), reported in Mega Watts (MW) for hyperparameter tuning and model comparison
- 70/10/20 split for train/validation/test data

RMSE measures the square root of the average of the squared differences between predicted and actual values. By squaring the errors before averaging, RMSE places greater emphasis on larger errors. MAE, on the other hand, calculates the average of the absolute differences between predicted and actual values. Unlike RMSE, MAE treats all errors equally, providing a more balanced measure of overall model accuracy without disproportionately penalising large deviations. Utilizing both would provide a balanced view of model performance.

It is also important to note for time-series forecasting, the validation approach differs from other validation settings. Traditional validation or cross validation involves randomly partitioning the dataset without considering the order of the data. In time-series however, to maintain temporal order, we do not shuffle the data, and the training set consists of only timesteps before the validation or test set with its chronological ordering preserved.

Together, these models provide a structured comparison across statistical, classical machine learning, and modern deep learning paradigms. This framework enables analysis not only of overall model accuracy, but also of how different modelling approaches scale with increasing prediction horizon, providing insight into their suitability for short-, medium-, and long-term forecasting tasks.

The model selection in this study is directly informed by the findings from the exploratory data analysis (Chapter 4). Specifically, the presence of strong multi-scale temporal structure motivates the use of models capable of capturing sequential dependencies. The non-linear relationship between temperature and demand supports the inclusion of flexible modelling approaches such as gradient boosting and neural networks. The observed lagged relationships and seasonal dependencies justify the use of autoregressive and sequence-based models. Finally, the presence of heteroscedasticity and increased error during peak demand periods highlights the need for models that are robust to extreme values and capable of modelling complex, regime-dependent behavior.

5.1.2 Classical Statistical SARIMAX

The SARIMAX model was used as a classical statistical baseline, representing traditional linear time-series forecasting methods. It incorporates both autoregressive and seasonal components, with temperature, weekend/weekday and 1-year demand lag included as an exogenous variable to capture external influences on electricity demand. Model parameters were selected based on standard time-series modelling practices, including identification of seasonal patterns, autocorrelation functions and residual analyses.

5.1.3 Gradient Boosting

To enhance predictive performance, a Gradient Boosting (GB) model was implemented as a representative machine learning approach capable of capturing non-linear relationships in the data. This model serves as a strong non-linear baseline commonly used for tabular time-series features. Explicit lag features (e.g. previous demand values at different time intervals) were constructed and used as inputs, enabling the model to learn complex interactions between past demand and exogenous variables such as temperature. The model was implemented using a tree-based ensemble method, with key hyperparameters including the number of estimators, learning rate, and maximum tree depth tuned to balance model complexity and predictive performance.

5.1.4 Long Short-Term Memory (LSTM) network

A Long Short-Term Memory (LSTM) network was employed to capture sequential dependencies in the electricity demand time series. LSTM models are particularly suitable for time series forecasting due to their ability to retain long-term temporal information through gated recurrent structures, allowing them to model complex temporal dynamics more effectively than traditional recurrent neural networks.

To examine the impact of increasing model complexity within deep learning approaches, multiple LSTM-based architectures were evaluated: a standard unidirectional LSTM, a Bidirectional LSTM (Bi-LSTM), and a Bi-LSTM augmented with an attention mechanism. The standard LSTM processes the input sequence in a forward temporal direction and serves as a baseline deep learning sequence model. The Bi-LSTM extends this architecture by processing the sequence in both forward and backward directions, enabling the model to capture richer contextual information from the input window.

To further enhance the model's ability to focus on the most relevant time steps, an attention mechanism was applied to the Bi-LSTM. This allows the model to assign different weights to historical observations when generating forecasts, improving its capacity to model non-uniform temporal dependencies. All LSTM variants were trained using sequences constructed from historical demand and exogenous variables, and their performance was evaluated to assess the impact of architectural complexity on forecasting accuracy across different horizons.

5.1.5 Transformer-based PatchTST model

Finally, a transformer-based model, PatchTST, was implemented to leverage recent advances in time series forecasting. PatchTST represents a state-of-the-art encoder-only transformer architecture designed specifically for long-horizon forecasting. Though there are newer transformers for time series released since, such as iTransformer (Liu et al., 2023) and TimeXer (Wang et al., 2024), PatchTST was the transformer first to deliver results better than the other family of models.

There were 2 key design changes. The first was it segments the input time series into fixed-length patches analogous to sliding windows in CNNs. Earlier transformers, such as Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021), and FEDformer (Zhou et al., 2022), input tokenization was primarily done pointwise. However, without capturing the neighbours of a particular timestep, much of the semantic information of the time series is lost, as pointwise measurements of a timestep do not give much meaning. PatchTST maximizes capturing such semantic information, by tokenizing entire patches made up of subsets of the time series.

The second is channel independence. The tokenization of multivariate inputs can happen on either a single-channel, or multiple channels. PatchTST uses the former configuration with motivation by the successes of CNNs. Nonetheless, self-attention mechanisms only apply to a single channel at a time. Different predictors are never directly interacted with each other except for sharing weights within

the transformer. However, this has shown to capture long-range temporal dependencies efficient enough and was an effective way of regularization.

The remainder of the transformer architecture is largely intact compared to the vanilla Transformer introduced by Vaswani et al. (2017), with only several smaller changes like using BatchNorm instead of LayerNorm.

5.2 Tuning of Hyper-parameters

To ensure consistency between the different model candidates, a single seed, 31415, was used for all family of models except SARIMAX (since its fitting process is deterministic). Though the criteria in selecting the best model are the RMSE and MAE, we find that there were multiple cases where smaller models were chosen because of faster training speeds without much of a trade-off in performance.

5.2.1 SARIMAX Tuning

The parameters of SARIMAX consists of the non-seasonal (p, d, q) and seasonal (P, D, Q, S) components. Both components have specifications for autoregressive (p / P), differencing (d / D) and moving average (q / Q) orders but the seasonal order specifically describes lag terms that occur in a fixed seasonality period (S). To find a starting point for the parameters, we followed the methodology outlined in Chapter 9 of Hyndman et al. (2021). In short, a heuristic is to utilize PACF and ACF from (Figures 4.5 to 4.7) to determine the autoregressive orders and moving average orders respectively. For differencing, we can refer to the trends from MSTL plot (Figure 4.12) to determine differencing orders to address non-stationarity.

The EDA section therefore suggests $p=3$ from the spikes at lags 1 to 3 in PACF and $P=Q=1$ with $S=48$ due to the spikes at lag-48 in PACF and ACF. Lastly, we used $D=1$ as some non-constant of the mean can be observed daily (likely due to weekday/weekends). Finally, a grid-search algorithm was used to investigate whether slight changes to these parameters can lead to better performance on the validation set.

order	seasonal_order	time_taken_seconds	training_size	validation_horizon	validation_rmse	rank_val_rmse
(5, 0, 0)	(1, 1, 1, 48)	262.28	1440	48	491.99	1
(3, 0, 0)	(1, 1, 1, 48)	200.96	1440	48	498.38	2
(4, 0, 0)	(1, 1, 1, 48)	217.99	1440	48	503.58	3
(4, 0, 0)	(1, 1, 1, 48)	233.37	1440	336	647.65	1
(3, 0, 0)	(1, 1, 1, 48)	225.33	1440	336	650.64	2
(5, 0, 0)	(1, 1, 1, 48)	272.79	1440	336	656.29	3
(4, 0, 0)	(1, 1, 1, 48)	225.56	1440	720	671.85	1
(3, 0, 0)	(1, 1, 1, 48)	218.88	1440	720	675.56	2
(5, 0, 0)	(1, 1, 1, 48)	154	720	720	682.88	3

Table 5-1 – The top 3 performance of various SARIMAX models grouped by forecast horizons.

Unlike the other machine learning models, fitting on more data did not improve SARIMAX's performance. In fact, we were not able to fit the entire training set into SARIMAX as it exceeded our available RAM. Nonetheless, we still compared different input sizes: 720, 1440, 2880 and 5760. These input sizes denote the number of timesteps before the last timestep in the training set. As shown in Table 5.1, a training size of 1440 observations yielded the best performance across all forecasting horizons. This is expected, as SARIMAX is fundamentally a linear model that relies heavily on the assumption of stationarity. Extending the training set further into the past introduces long-term macroscopic trends and shifting structural cycles that violate this assumption. Consequently, parameter estimation via Maximum Likelihood Estimation (MLE) ends up averaging these historical variations into the model's coefficients, making the final model less representative of the localized, recent dynamics found in the validation data.

In contrast to what was suggested by the PACF plot, order (3,0,0) never performed the best in any of the horizons. Model (4,0,0) had the best RMSE for 336 and 720-horizon. For 48-horizon, though (5,0,0) had a better RMSE, it took around 20% longer to fit for only a small improvement over (3,0,0) and (4,0,0). Thus, for consistency, we chose (4,0,0) to represent all 3 horizons.

As SARIMAX models are estimated through the maximum likelihood estimation with the assumed underlying distribution as Normal, we check whether our fitted model satisfies the required assumptions. Specifically, we require standardized residuals to have a constant variance, centered on zero and be Normally and independently distributed. Figure 5.1a below shows the diagnostic plots for the fitted model. We can see the mean is approximately zero with a constant variance from the line plot and histogram. The histogram and Q-Q plot also resemble a Normal distribution, though there is still a slight skew to the right. We also see from the autocorrelation plot that the lags of the residuals are close to zero, implying independence. Finally, Figure 5.1b confirms many of the same observations mentioned, supported by the failure to reject H_0 on Ljung-Box, Heteroskedasticity and Jarque-Bera tests.

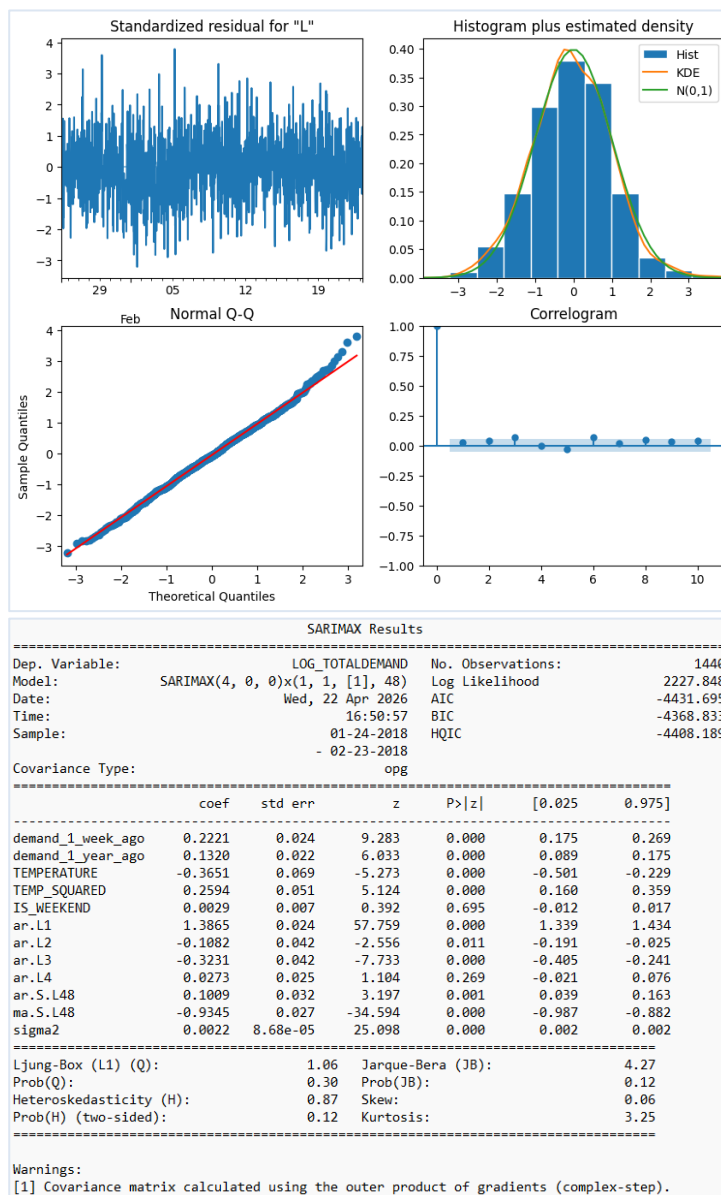


Figure 5-1 a) Top, SARIMAX residual diagnostic plots. b) Bottom, model summary of the best model.

5.2.2 Gradient Boosting

Unlike the other models, Gradient Boosting did not have a context window, nor did its performance improve by only training on recent data, like in SARIMAX's case. For Gradient Boosting, we directly utilised a grid search algorithm to optimize the number of estimators, maximum depth and learning rate. Early experimentation on number of estimators quickly ruled out 500 or below as their performance was subpar. Hence, the remaining search space consisted of:

- Number of estimators: 600, 1000 and 1200
- Maximum Depth: 3 and 5
- Learning Rate: 0.1, 0.05 and 0.01

Altogether, this meant 18 models were tested for our three forecast horizons. Table 5.2 shows the RMSE of all the top 3 models were very close, with the horizon 720 only have 1.49 RMSE or 0.27% difference. Depth greater than 5 seems to be worse in all cases except for horizon 48. This may suggest Gradient Boosting is sensitive to overfitting on adding depth, especially for longer forecast horizons. Also, all other learning rates besides 0.1 did not improve performance as well.

horizon	n_estimators	max_depth	learning_rate	rmse	rank_val_rmse
48	1200	5	0.1	382.25	1
48	1000	3	0.1	385.54	2
48	600	3	0.1	388.85	3
336	1200	3	0.1	513.04	1
336	1000	3	0.1	515.18	2
336	600	3	0.1	516.81	3
720	1000	3	0.1	550.9	1
720	1200	3	0.1	551.74	2
720	600	3	0.1	552.39	3

Table 5-2: Top 3 models by forecast horizon

Introducing additional estimators resulted in only marginal performance gains relative to increased computational costs, indicating diminishing returns on this aspect. Therefore, following Occam's Razor principle, we decided to opt for the smallest model displayed in above table for all three forecast horizons. Specifically, the number of estimators was set to 600, with max depth of 3 and a learning rate of 0.1.

5.2.3 LSTM

For the LSTM-based models, hyperparameters governing both model capacity and training dynamics were systematically tuned to optimise performance across multiple forecasting horizons. The tuning process aimed to balance model expressiveness with generalisation, particularly for longer-term predictions where overfitting and instability are more pronounced.

Key architectural hyperparameters included the hidden size and number of layers. The hidden size determines the dimensionality of the learned temporal representation, with larger values enabling the model to capture more complex patterns, but increasing the risk of overfitting. The number of layers was varied to assess whether deeper architectures could better capture hierarchical temporal features. While deeper models offer greater representational power, they also increase training complexity and sensitivity to hyperparameter selection.

Dropout was incorporated as a regularisation mechanism to mitigate overfitting, particularly in deeper models. Different dropout rates were evaluated to determine the optimal balance between model robustness and the retention of meaningful temporal information.

Training-related hyperparameters were also tuned. The learning rate was adjusted to ensure stable convergence, with lower values improving stability but increasing computational time. Early stopping was implemented using a patience parameter to prevent overfitting, terminating training when validation performance ceased to improve.

In addition to hyperparameter tuning, multiple LSTM architectures were evaluated, including a standard unidirectional LSTM, a bidirectional LSTM (Bi-LSTM), and a Bi-LSTM augmented with an attention mechanism. The unidirectional LSTM serves as a baseline sequential model, while the Bi-LSTM captures both forward and backward temporal dependencies within the input sequence. The attention-based model further extends this capability by allowing the network to assign varying importance to different time steps, which is particularly beneficial for modelling non-uniform temporal relationships.

Hyperparameter tuning for the LSTM models was guided by validation performance, with the objective of minimising validation loss as a proxy for generalisation accuracy. Rather than selecting configurations based solely on training performance, all tuning decisions were made using validation loss to ensure that the model remained robust to unseen data. Early stopping was employed to prevent overfitting, with the optimal model defined as the epoch achieving the lowest validation loss. This approach ensures that selected configurations balance model capacity with generalisation performance, particularly for longer forecasting horizons where overfitting is more pronounced.

Ultimately, Bi-LSTM prevailed over the other architectures. From Table 5.3, the smaller model was chosen as the best model.

horizon	lookback	hidden_size	num_layers	dropout	num_attention_heads	val_loss	val_rmse
48	168	128	2	0.2	4	0.11	457.67
48	168	256	4	0.35	8	0.13	477.78
336	336	128	2	0.2	4	0.25	685.78
336	336	256	4	0.35	8	0.30	738.06
720	720	128	2	0.2	4	0.34	791.05
720	720	256	4	0.35	8	0.35	797.33

Table 5-3. Performance of various LSTM models grouped by forecast horizons

5.2.4 Transformer-based PatchTST model

For the PatchTST model, we referred to the original hyperparameters discussed by Nie et al. (2023). Specifically, we chose PatchTST/42 as a starting point for tuning. We then created another “larger” model with greater dimensions of the embedding, feed-forward layer in the encoder layers, and greater number of encoder layers. We refer PatchTST/42 as the “small” model. We then compared the two models’ validation RMSE with 2 sets of learning rates and dropout percentage as seen in Table 5.4 below. All models were trained with 10 epochs as patience for early stopping.

horizon	lookback	d_model	n_enc_layers	dim_ff	dropout	lr	val_rmse	time_taken_seconds	rank
48	336	512	5	2048	0	5.00E-06	406.14	206.4	1
48	720	128	3	256	0	5.00E-06	411.97	63.13	2
48	336	128	3	256	0	5.00E-06	413.64	48.54	3
48	720	512	5	2048	0	5.00E-06	414.71	436.48	4
336	336	512	5	2048	0	5.00E-06	597.3	207.74	1
336	720	512	5	2048	0	5.00E-06	602.79	441.29	2
336	720	128	3	256	0	5.00E-06	604.32	64.53	3
336	336	128	3	256	0	5.00E-06	609.56	48.75	4
720	336	128	3	256	0	5.00E-06	634.88	50.39	1
720	336	512	5	2048	0	5.00E-06	634.96	211.5	2
720	720	128	3	256	0	5.00E-06	642.92	66.84	3
720	720	512	5	2048	0	5.00E-06	648.35	447.51	4

Table 5-4: Performance of various PatchTST models grouped by forecast horizons

Shown in the appendix section 10.1.1, models with 20% dropout and 0.0001 learning rates had worse performances compared to no dropout and smaller learning rate. Table 5.4 shows the remaining models. The “small” model is the one with embedding dimension ($d_{\text{model}}=128$), 16 attention heads, and 3 encoder layers.

For horizons 48 and 336, the “large” model with 512 d_{model} and 336 lookback windows performed the best with validation RMSE 406.14 and 597.3 respectively. The closest “small” model are the ones with 128 d_{model} and 720 lookback windows, scoring 411.97 and 604.32 validation RMSE respectively. However, it is crucial to note the larger models take around 3-times longer to train per epoch while being around 25 times larger in terms of number of parameters (small models has 663,091 while the large model has 16,824,373 parameters). From a practical perspective, taking 3-times longer to train for a 1.2% to 1.4% decrease in RMSE is not worth it. Thus, we choose the “small” models with 720 context length, 0% dropout and learning rate 5.00E-06 as our best models for these two forecast horizons.

Horizon 720 is more straightforward, with the 336 lookback smaller model being as good, if not better, than the larger models. Therefore, we use this as the best model for this forecast horizon. Note this is basically the original configuration of PatchTST\42 by Nie et al. (2023) but with 0% dropout.

We also investigated to see whether the presence of additional predictors really contributes to performance. Since predictors never directly interact with each other by the attention mechanism, but only through shared weights, a comparison with and without predictors can potentially give us insights on how much exogenous variables affect the outcome. Table 5.5 shows the best model fitted without any predictors.

horizon	lookback	d_model	dropout	learning_rate	validation_rmse	time_taken_seconds
48	720	128	0	5.00E-06	413.36	51.67
48	336	128	0	5.00E-06	417.01	48.95
336	720	128	0	5.00E-06	609.27	51.95
336	336	128	0	5.00E-06	615.93	50.58
720	336	128	0	5.00E-06	644.78	50.04
720	720	128	0	5.00E-06	660.35	50.91

Table 5-5: Performance of univariate PatchTST models grouped by forecast horizons.

We see there is a very small increase in RMSE. Around less than or nearly 1% increase for horizons 48 and 336, and 1.5 % to 2.7% for horizon 720. This suggests either channel independence has a powerful effect on regularization, or modelling just on the demand variable is able to cover much of the information that temperature would have had provided in the case of PatchTST.

5.3 Model Training Behaviour

Before evaluating forecasting performance, the training behaviour of the models is analysed to provide insight into convergence, stability, and generalisation.

The training behaviour of the SARIMAX model differs from that of neural network-based approaches, as it does not involve iterative learning across epochs. Instead, model fitting is performed through maximum likelihood estimation, with convergence determined by optimisation of the likelihood function. Furthermore, as previously noted in section 5.2.1, unlike deep learning architectures that typically scale well with data, incorporating longer historical training data introduces non-stationary trends that naturally distort its linear parameters during MLE fitting. Thus, SARIMAX performed optimally with a restricted training window (1440 steps).

Gradient Boosting models are trained iteratively, with each successive tree attempting to correct the errors of the previous ensemble. In contrast to the highly variable training dynamics often seen in neural networks, Gradient Boosting demonstrated smooth and stable convergence across configurations, showing no signs of abrupt divergence or instability. However, while its training behaviour is highly stable, its reliance on fixed lagged inputs ultimately limits its ability to adapt to long-range temporal dependencies across longer forecasting horizons compared to our deep learning models.

Neural network-based models exhibit more complex training dynamics. Representative training and validation loss curves for both LSTM and PatchTST models are shown in Figures 5.2–5.3 for different combinations of lookback window and forecasting horizon.

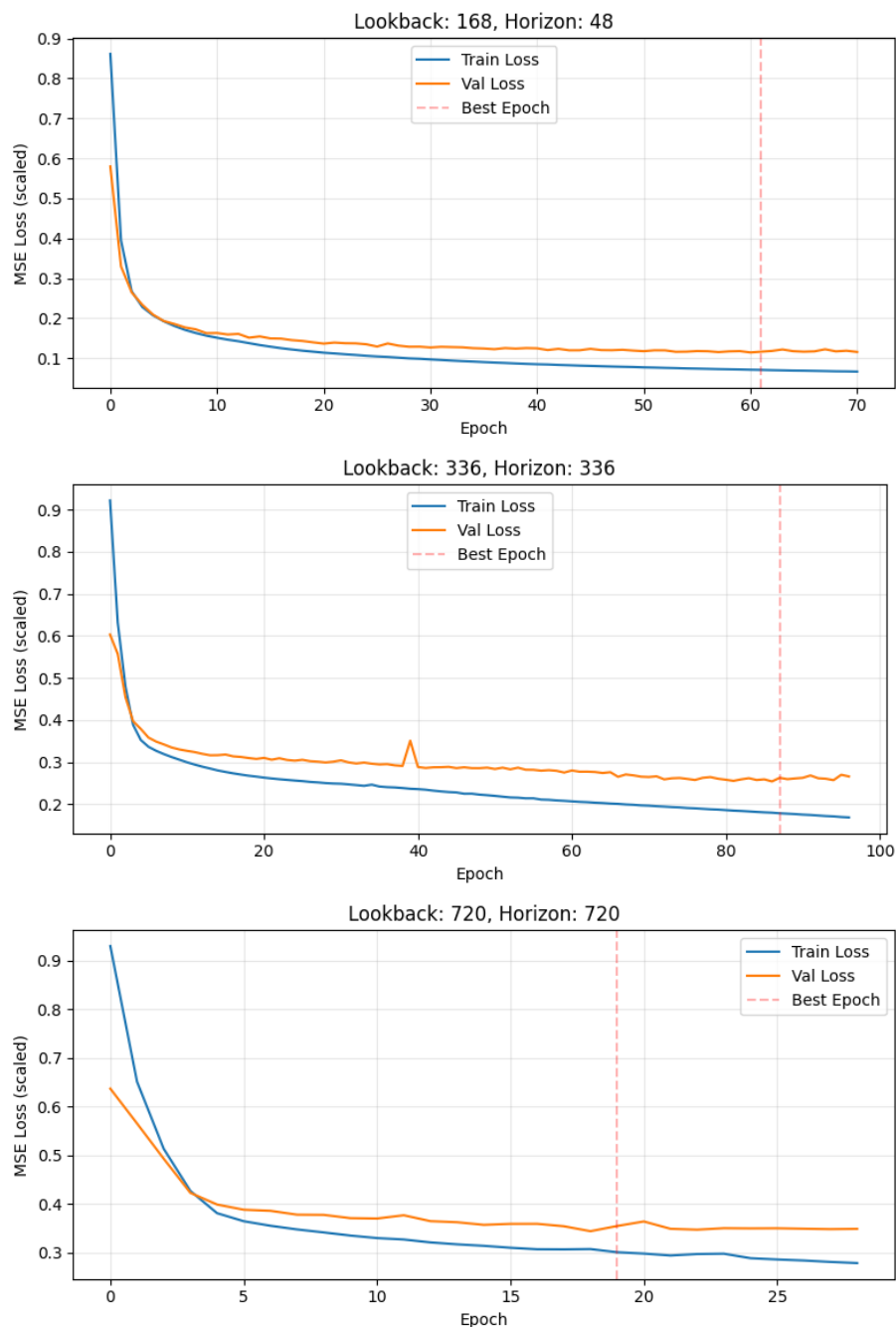


Figure 5-2: LSTM training and validation loss curves

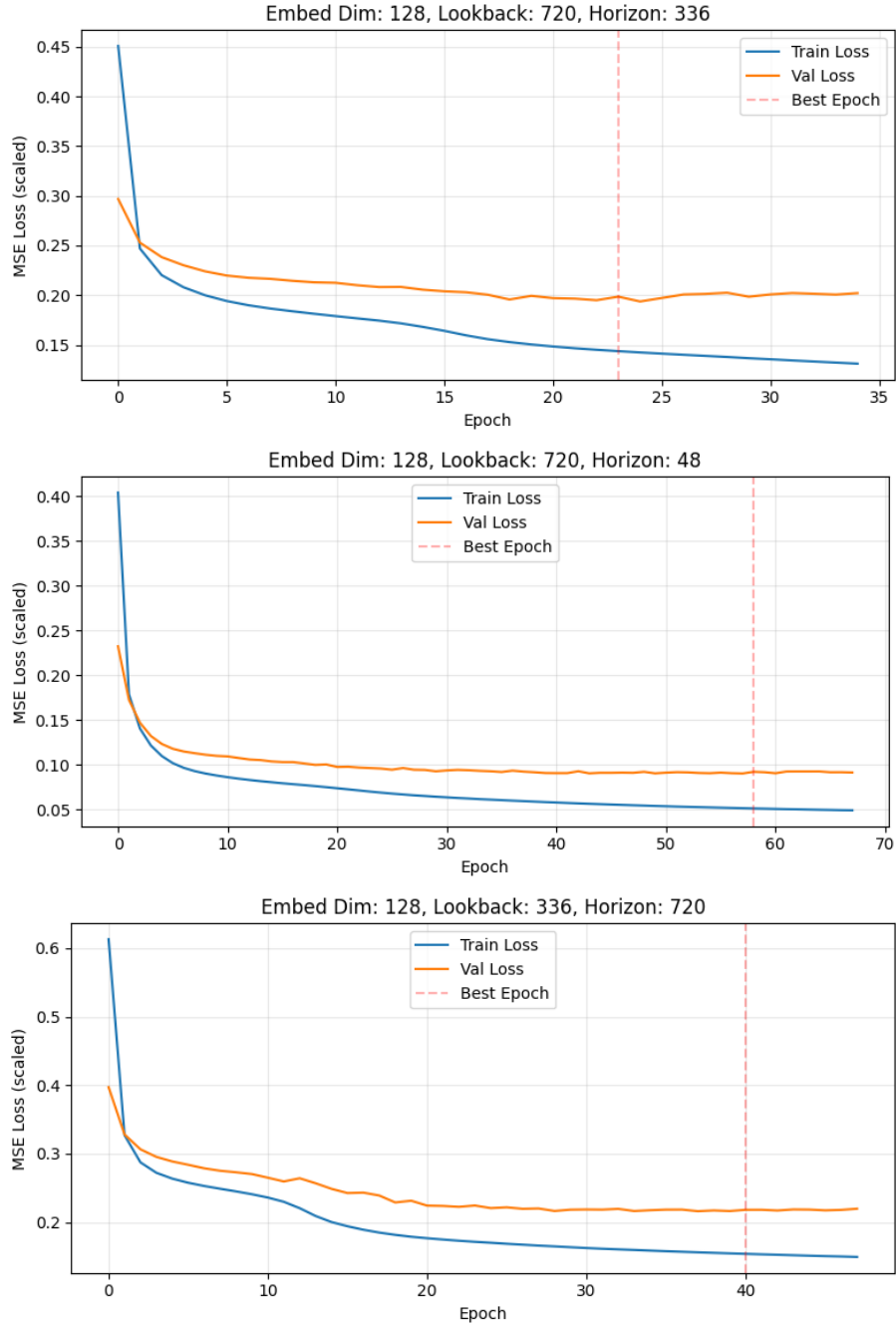


Figure 5-3: PatchTST training and validation loss curves

Across all configurations, both models demonstrate rapid initial learning, with a sharp reduction in loss during the first few epochs. This indicates that the models are able to quickly capture dominant temporal patterns in the data. However, the subsequent training dynamics differ between architectures and forecasting setups.

For the LSTM-based models, training loss continues to decrease steadily throughout training, while validation loss typically plateaus after a relatively small number of epochs. This results in a persistent gap between training and validation loss, indicating moderate overfitting. The effect becomes more pronounced as the forecasting horizon increases. For shorter horizons (e.g. 48 steps), validation loss stabilises at a lower level and remains relatively smooth. In contrast, for longer horizons (e.g. 336 and 720 steps), validation loss is higher and exhibits greater fluctuation, reflecting increased prediction difficulty and reduced generalisation capacity.

The application of early stopping is therefore critical for LSTM models. In most cases, the optimal model corresponds to an early epoch where validation loss reaches its minimum before diverging from the training curve. Beyond this point, continued training yields diminishing returns and may degrade generalisation performance.

The PatchTST models exhibit similar initial convergence behaviour, but with notable differences in generalisation. While training loss decreases in a comparable manner, the gap between training and validation loss is generally smaller than that observed in LSTM models, indicating improved generalisation. Validation loss tends to stabilise earlier and at lower values, particularly for short- and medium-horizon forecasts.

Differences in behaviour are also observed across input configurations. Models using longer lookback windows do not consistently produce lower validation loss, particularly for longer forecasting horizons. In some cases, shorter lookback windows converge more efficiently and achieve comparable or improved validation performance, suggesting that additional historical input may introduce noise rather than useful signal.

Overall, these results highlight key differences in how the models learn temporal structure. SARIMAX exhibits stable but inflexible behaviour, Gradient Boosting shows efficient but feature-limited learning, LSTM models capture short-term dependencies but suffer from increasing overfitting at longer horizons, and PatchTST models demonstrate more stable convergence and improved generalisation.

6 Results

6.1 Summary of Results

The final performance results for all models across forecasting horizons are summarised in Table 6.1. A clear trend emerges in model performance as the prediction horizon increases. Note results in Table 6.1 are the mean values of RMSE/MAE metrics generated by 6 random seed values: 31415, 27182, 14142, 17320, 22360 and 57721. Appendix section 10.1.3 shows the same table with standard deviation.

Overall, these results highlight a clear trade-off between model complexity and forecasting horizon. PatchTST provides the most consistent and accurate performance for short- and medium-term forecasting, while simpler models such as SARIMAX remain competitive at longer horizons. Gradient Boosting offers stable performance across all horizons but does not achieve the best accuracy in any regime. LSTM models, while effective for short-term temporal modelling, struggle to generalise as horizon increases.

	Horizon - 48 steps		Horizon - 336 steps		Horizon - 720 steps	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
<i>SARIMAX</i>	544.65	367.25	650.79	456.09	667.58	474.72
<i>Gradient Boosting</i>	536.34	380.49	655.67	471.72	690.82	498.17
<i>LSTM</i>	564.98	402.08	790.94	574.38	858.99	635.97
<i>PatchTST</i>	468.07	308.63	637.35	445.23	694.99	496.41

Table 6-1

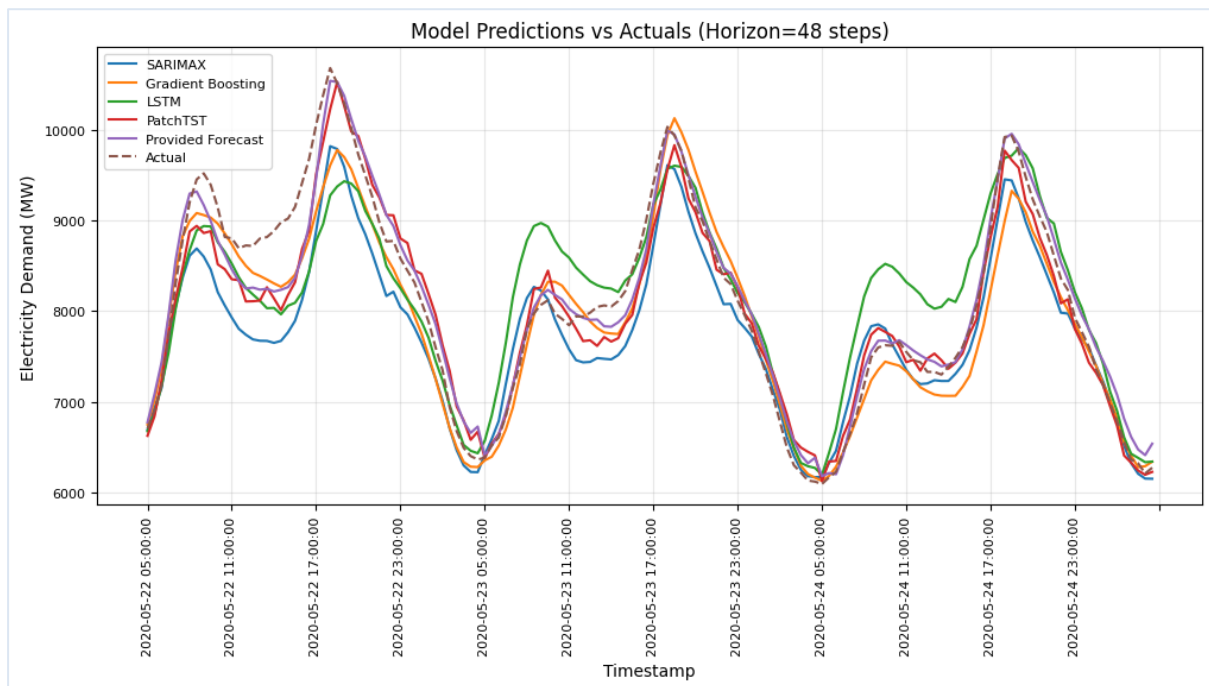


Figure 6-1. Plot comparing the different models for the 48-step horizon.

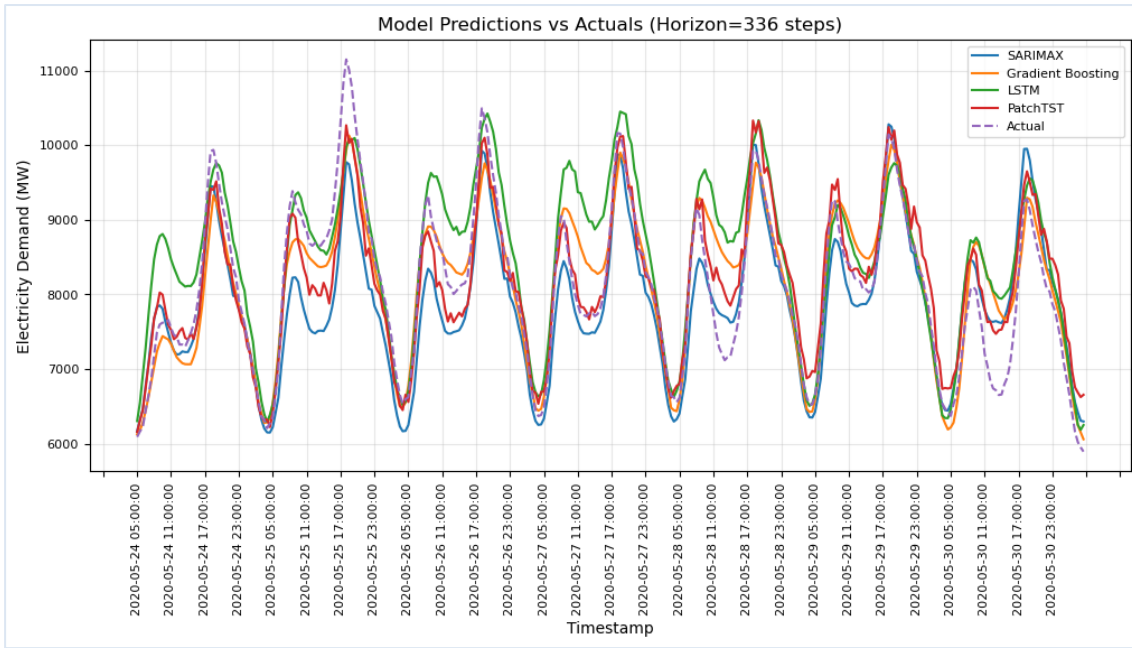


Figure 6-2. Plot comparing the different models for the 336-step horizon.

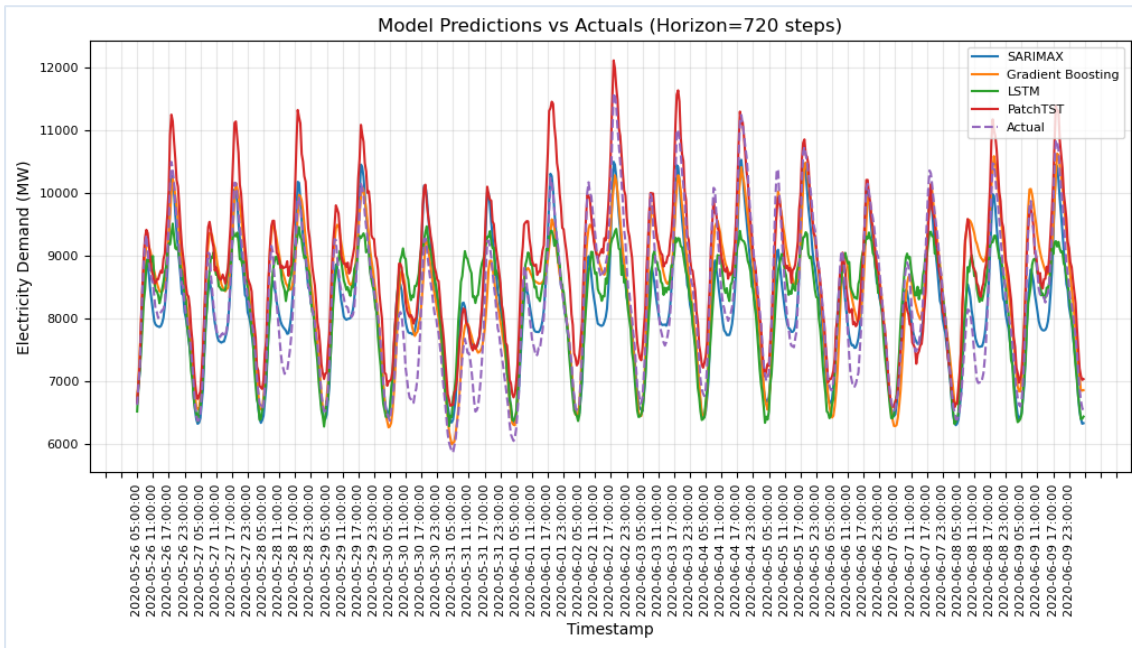


Figure 6-3. Plot comparing the different models for the 720-step horizon.

6.1.1 Results for 48-step Forecast (Short-term)

For short-term forecasting ($h = 48$), PatchTST significantly outperforms the others (RMSE 468.07, MAE 308.63), demonstrating the effectiveness of transformer architectures in capturing short-term temporal dependencies. Gradient Boosting (RMSE 536.64) and SARIMAX (RMSE 544.65) follow closely, proving that both tree-based and linear autoregressive models remain highly competitive where strong temporal persistence exists. Conversely, LSTM produces the highest error (RMSE 564.98), suggesting its complexity and sequential learning offer limited benefits and may lead to overfitting in this regime. Figure 6.1 shows a model comparison of a 48-step prediction horizon.

6.1.2 Results for 336-step Forecast (Medium-term)

At the medium-term horizon ($h = 336$), error increases across all models as the performance gap

narrows. PatchTST retains a clear advantage (RMSE 637.35, MAE 445.23), proving its ability to capture longer-range dependencies. This performance is consistent with the training behaviour observed in Section 6.2, where PatchTST demonstrated improved generalisation and stable convergence. SARIMAX (RMSE 650.79) and Gradient Boosting (RMSE 655.67) yield comparable predictive power, effectively capturing dominant patterns but struggling with more complex temporal relationships. Meanwhile, LSTM experiences a sharp decline in performance (RMSE 790.94, MAE 574.38), highlighting its difficulty in maintaining generalization over longer sequences. This behaviour is also consistent with the training dynamics observed earlier, where validation loss plateaued early and exhibited increased instability for longer horizons. Figure 6.2 shows a model comparison of a 336-step prediction horizon.

6.1.3 Results for 720-step Forecast (Long-term)

For the long-term forecasting horizon ($h = 720$), all models exhibit a further degradation in performance, reflecting the significant uncertainty associated with extended prediction intervals. At this horizon, differences between models become less pronounced, indicating that long-range forecasting is inherently constrained by the limited predictive information available in historical data. Here, the simpler SARIMAX model marginally achieves the lowest error (RMSE 667.58) suggesting that capturing dominant seasonal structure becomes more valuable than modelling fine temporal patterns. Gradient Boosting (RMSE 690.82) and PatchTST (RMSE 694.99, MAE 496.41) perform similarly, indicating that the advantages of complex, attention-based, or non-linear modelling diminish at extended horizons. LSTM continues to exhibit the poorest performance (RMSE 858.99, MAE 635.97), suffering from extreme instability and overfitting in long-range predictions. Figure 6.3 shows a model comparison of a 720-step prediction horizon.

7 Discussion

This study evaluated a range of forecasting models for electricity demand across varying horizons, revealing a clear trade-off between model complexity and forecast distance. At shorter horizons, PatchTST to demonstrate clear superiority indicating that transformer-based architectures are highly effective at capturing and modelling short-term temporal dependencies. However, as the horizon extends, the predictive signal available in historical data for this horizon degrades. Consequently, the performance gap narrows, and complex models like PatchTST and Gradient Boosting lose their distinct advantage. By the long-term horizon ($h = 720$), the forecasting task becomes increasingly dominated by uncertainty. In this regime, capturing broad seasonal structure appears more important than modelling fine-grained temporal relationships.

An important and somewhat surprising observation is the continued relevance of classical statistical models. While Nie et al. (2023) demonstrated that PatchTST effectively outperforms simple linear models (such as DLinear) in long-term forecasting, our study shows that the reality is more complex. By bridging the gap in comparative literature between modern transformers and classical statistical baselines, we found that SARIMAX unexpectedly outperformed PatchTST at the longest horizon. This suggests that while transformer architectures beat simple linear machine learning models, classical autoregressive models with explicit seasonal components (SARIMAX) may be more robust under conditions of high uncertainty, where overfitting and noise amplification become significant challenges for more complex architectures. In other words, predictive signal available in historical data becomes insufficient to support highly accurate forecasts in longer horizons, regardless of model sophistication, highlighting an important trade-off between model complexity and forecasting horizon.

The results also reinforce the importance of temporal structure identified in the exploratory analysis. Strong short-term autocorrelation and clear seasonal patterns support the effectiveness of both statistical and machine learning models. However, the multi-scale nature of the data introduces challenges for models that rely on sequential processing, such as LSTM. In contrast, attention-based models are better suited to capturing complex temporal relationships, particularly at intermediate horizons.

The relatively poor performance of the LSTM model across medium- and long-term horizons highlights the limitations of recurrent architectures for this task. Despite their theoretical ability to model sequential dependencies, LSTMs struggle to retain information over extended sequences and are prone to overfitting, as observed in the training behaviour. This finding is consistent with the literature by Li et al. (2022), where LSTM-based models performed poorly in all horizons (refer to section 10.1.2 in the appendix).

Finally, the results emphasise that improvements in long-horizon forecasting may depend less on increasing model complexity and more on incorporating additional predictive information. In particular, the inclusion of exogenous variables such as weather forecasts or real-time system data may be necessary to achieve meaningful gains in long-term predictive accuracy.

8 Conclusion and Further Issues

8.1 Summary of Study

This study compared statistical, machine learning, and deep learning approaches for electricity demand forecasting. Specifically, SARIMAX, Gradient Boosting, LSTM, and PatchTST models were evaluated across multiple forecasting horizons to assess their ability to capture temporal dependencies and generalise to unseen data.

Model performance was evaluated using RMSE and MAE, with additional analysis of training behaviour to understand convergence, stability, and generalisation characteristics. The models were also assessed relative to a provided operational forecast to provide context for real-world performance.

8.2 Key Findings

The results demonstrate that model performance is strongly dependent on forecasting horizon, with no single model consistently outperforming others across all regimes.

For short-term forecasting, the PatchTST model achieves the best performance, with Gradient Boosting and SARIMAX also producing competitive results. At this horizon, strong temporal persistence allows multiple modelling approaches to perform effectively.

At the medium-term horizon, PatchTST maintains the lowest error, indicating its ability to capture longer-range temporal dependencies. However, the performance gap between models begins to narrow, with SARIMAX and Gradient Boosting producing similar results.

For long-term forecasting, all models exhibit substantial performance degradation, and differences between models become less pronounced. In this regime, SARIMAX achieves the lowest error, suggesting that simpler models remain competitive when predictive uncertainty is high.

Across all horizons, the LSTM model consistently underperforms relative to other approaches, with performance deteriorating significantly as horizon increases. This highlights the limitations of recurrent architectures for long-range forecasting tasks.

8.3 Implications

These findings highlight that forecasting performance is not solely determined by model complexity, but by how well a model aligns with the forecasting horizon and underlying data characteristics.

Advanced models, such as transformer-based architectures, provide clear benefits for short- and medium-term forecasting by capturing complex temporal dependencies. However, for long-horizon forecasting, the advantages of increased model complexity diminish, and performance becomes increasingly dominated by uncertainty.

The continued competitiveness of SARIMAX at longer horizons suggests that simpler statistical models may offer greater robustness under conditions where predictive signal is weak. This challenges the assumption that more complex models will always yield superior performance.

Overall, effective forecasting requires careful consideration of both model selection and forecasting horizon, rather than reliance on a single modelling approach.

An important consideration is the role of input data in determining model performance. Deep learning models, particularly transformer-based architectures, are designed to capture complex and high-dimensional relationships and typically benefit from large and diverse input datasets. In this study, the models rely primarily on historical demand and limited exogenous information, which may restrict their ability to fully utilise their representational capacity.

This limitation becomes more pronounced at longer forecasting horizons, where predictive performance depends on the availability of informative forward-looking signals. In the absence of such inputs, even advanced models are constrained by the information content of the data, leading to convergence in performance across modelling approaches. This suggests that the relatively modest gains observed for deep learning models, particularly at longer horizons, may reflect limitations in data richness rather than inherent shortcomings of the architectures themselves.

8.4 Future Work

Future work should focus on improving model performance through the incorporation of additional predictive information. In particular, the inclusion of exogenous variables such as weather forecasts, calendar effects, and real-time system data may provide significant improvements, especially for short- and medium-term forecasting.

Further optimisation of model architectures and hyperparameters may also yield incremental performance gains, particularly for transformer-based models. In addition, exploring hybrid approaches that combine statistical and machine learning methods may provide a balance between robustness and flexibility.

Finally, extending the analysis to additional regions or datasets would allow for assessment of model generalisability and robustness across different demand profiles and operating conditions.

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Appendix

9 Codes

https://github.com/laukamkit/capstone_project_GroupA

10 Appendix

10.1.1 All PatchTST Model Candidate Performance

horizon	lookback	d_model	n_enc_layers	dim_ff	dropout	lr	val_rmse	time_taken_seconds	rank
48	336	512	5	2048	0	5.00E-06	406.14	206.4	1
48	720	128	3	256	0	5.00E-06	411.97	63.13	2
48	336	128	3	256	0	5.00E-06	413.64	48.54	3
48	720	512	5	2048	0	5.00E-06	414.71	436.48	4
48	336	128	3	256	0.2	0.0001	460.45	48.62	5
48	336	512	5	2048	0.2	0.0001	461.2	211.94	6
48	720	128	3	256	0.2	0.0001	461.23	64.94	7
48	720	512	5	2048	0.2	0.0001	470.61	448.61	8
336	336	512	5	2048	0	5.00E-06	597.3	207.74	1
336	720	512	5	2048	0	5.00E-06	602.79	441.29	2
336	720	128	3	256	0	5.00E-06	604.32	64.53	3
336	336	128	3	256	0	5.00E-06	609.56	48.75	4
336	336	512	5	2048	0.2	0.0001	632.83	213.45	5
336	336	128	3	256	0.2	0.0001	640.48	50.98	6
336	720	128	3	256	0.2	0.0001	648.63	65.56	7
336	720	512	5	2048	0.2	0.0001	664.69	453.43	8
720	336	128	3	256	0	5.00E-06	634.88	50.39	1
720	336	512	5	2048	0	5.00E-06	634.96	211.5	2
720	720	128	3	256	0	5.00E-06	642.92	66.84	3
720	720	512	5	2048	0	5.00E-06	648.35	447.51	4
720	336	512	5	2048	0.2	0.0001	671.7	217.2	5
720	720	128	3	256	0.2	0.0001	672.81	67.88	6
720	336	128	3	256	0.2	0.0001	673.36	53.56	7
720	720	512	5	2048	0.2	0.0001	692.7	460.1	8

Appendix

10.1.2 LSTM Performance Table on ECL Dataset by Li et al. (2022)

Table 1: Multivariate long sequence time-series forecasting results on the electricity-field datasets.																					
Dataset		ETTh1					ETTh2					ETTm1					ECL				
Method	Metric	24	48	168	336	720	24	48	168	336	720	24	48	96	288	672	48	168	336	720	960
LSTNet [5]	MSE	1.175	1.344	1.865	2.477	1.925	2.632	3.487	1.442	1.372	2.403	1.856	1.909	2.654	1.009	1.681	0.279	0.318	0.357	0.442	0.473
	MAE	0.793	0.864	1.092	1.193	1.084	1.337	1.577	2.389	2.429	3.403	1.058	1.085	1.378	1.902	2.701	0.337	0.368	0.391	0.433	0.443
LSTMa [1]	MSE	0.536	0.616	1.058	1.152	1.682	1.049	1.331	3.987	3.276	3.711	0.621	1.392	1.339	1.740	2.736	0.388	0.492	0.778	1.528	1.343
	MAE	0.528	0.577	0.725	0.794	1.018	0.689	0.805	1.560	1.375	1.520	0.629	0.939	0.913	1.124	1.555	0.444	0.498	0.629	0.945	0.886
Reformer [4]	MSE	0.887	1.159	1.686	1.919	2.177	1.381	1.715	4.484	3.798	5.111	0.598	0.952	1.267	1.632	1.943	1.312	1.453	1.507	1.883	1.973
	MAE	0.630	0.750	0.996	1.090	1.218	1.475	1.585	1.650	1.508	1.793	0.489	0.645	0.795	0.886	1.006	0.911	0.975	0.978	1.002	1.185
LogTrans [6]	MSE	0.656	0.670	0.888	0.942	1.109	0.726	1.728	3.944	3.711	2.817	0.341	0.495	0.674	1.728	1.865	0.267	0.290	0.305	0.311	0.333
	MAE	0.600	0.611	0.766	0.766	0.843	0.638	0.944	1.573	1.587	1.356	0.495	0.527	0.674	1.656	1.721	0.366	0.382	0.395	0.397	0.413
Informer [13]	MSE	0.509	0.551	0.878	0.884	0.941	0.446	0.934	1.512	1.665	2.340	0.325	0.472	0.642	1.219	1.651	0.269	0.300	0.311	0.308	0.328
	MAE	0.523	0.563	0.722	0.753	0.768	0.523	0.733	0.996	1.035	1.209	0.440	0.537	0.626	0.871	1.002	0.351	0.376	0.385	0.385	0.406
TCN* [7]	MSE	0.467	0.478	0.652	0.787	0.927	0.424	0.594	2.355	2.396	2.572	0.210	0.219	0.243	0.724	2.136	-	-	-	-	-
	MAE	0.477	0.492	0.606	0.675	0.770	0.465	0.589	1.168	1.214	1.241	0.280	0.316	0.343	0.633	1.211	-	-	-	-	-
SCINet [7]	MSE	0.341	0.368	0.451	0.502	0.583	0.188	0.279	0.505	0.618	1.074	0.126	0.169	0.191	0.365	0.713	-	-	-	-	-
	MAE	0.379	0.395	0.457	0.497	0.560	0.288	0.358	0.504	0.560	0.761	0.229	0.274	0.291	0.415	0.604	-	-	-	-	-
SNaive	MSE	0.424	0.465	0.656	0.683	0.650	0.266	0.322	0.578	0.567	0.550	0.168	0.168	0.188	0.225	0.282	0.212	0.212	0.212	0.228	0.267
	MAE	0.389	0.407	0.509	0.520	0.518	0.304	0.339	0.483	0.491	0.488	0.269	0.269	0.287	0.315	0.353	0.278	0.278	0.279	0.295	0.325
Linear	MSE	0.305	0.341	0.404	0.434	0.490	0.166	0.218	0.323	0.386	0.665	0.090	0.121	0.162	0.261	0.364	0.116	0.145	0.163	0.194	0.212
	MAE	0.357	0.379	0.422	0.445	0.501	0.261	0.299	0.374	0.419	0.549	0.185	0.219	0.253	0.325	0.394	0.213	0.240	0.260	0.291	0.308

10.1.3 Results with 6 Different Random Seeds

	Horizon - 48 steps		Horizon - 336 steps		Horizon - 720 steps	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
SARIMAX	544.65	367.25	650.79	456.09	667.58	474.72
Gradient Boosting	536.34±0.01	380.49±0.01	655.67±0.1	471.72±0.2	690.82±0.2	498.17±0.2
LSTM	564.98±20.9	402.08±13.5	790.94±17.1	574.38±16.9	858.99 ±13.2	635.97±10.6
PatchTST	468.07±3.9	308.63±2.7	637.35±15.0	445.23±15.0	694.99±19.4	496.41±18.4

Same as Table 6.1 with standard deviation displayed after the ± symbol